

# Comparison Between RS67 Vs RS57-70

Chatura Kuruppu

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## Slide 02: Input Files (Roadset & Target Order)

- merged\_RS57\_LH2\_1\_1138.root
- merged\_RS57\_LD2\_3\_1138.root
- merged\_RS57\_Empty\_2\_1138.root
- merged\_RS59\_LH2\_1\_465.root
- merged\_RS59\_LD2\_3\_466.root
- merged\_RS59\_Empty\_2\_466.root
- merged\_RS62\_LH2\_1\_1234.root
- merged\_RS62\_LD2\_3\_1237.root
- merged\_RS62\_Empty\_2\_1234.root
- merged\_RS67\_3089LH2.root
- merged\_RS67\_3089LD2.root
- merged\_RS67\_3089flask.root
- merged\_RS70\_LH2\_1\_264.root
- merged\_RS70\_LD2\_3\_266.root
- merged\_RS70\_Empty\_2\_267.root

**Remark:** Currently we don't process RS5a, RS5b files.

## Slide 03: Physics Constants & Normalizations

| Parameter                                               | Value                         |
|---------------------------------------------------------|-------------------------------|
| PoT LH <sub>2</sub>                                     | $2.7639 \times 10^{17}$       |
| PoT LD <sub>2</sub>                                     | $1.3197 \times 10^{17}$       |
| PoT Flask                                               | $5.5905 \times 10^{16}$       |
| LH <sub>2</sub> Target Density                          | 3.5966 mol/cm <sup>2</sup>    |
| LD <sub>2</sub> Target Density                          | 8.0431 mol/cm <sup>2</sup>    |
| LH <sub>2</sub> / LD <sub>2</sub> Target Length         | 50.8 cm                       |
| Avogadro Constant                                       | $6.022 \times 10^{23}$        |
| xF Bin Width                                            | 0.05                          |
| Nuclear Int. Length (LH <sub>2</sub> /LD <sub>2</sub> ) | 52.0 / 54.7 g/cm <sup>2</sup> |

## Slide 04: Event Selection Summary

**Dimuon Level**  $|dx| < 0.25$ ,  $\text{mass} \in [4.2, 8.8]$ ,  $x_F \in [-0.1, 0.95]$ ,  $x_T \in [0.05, 0.58]$ ,  $\chi^2 < 18$ .

**Track Level**  $\chi^2_{\text{target}} < 15$ ,  $p_{z_{st1}} \in [9, 75]$ , Hits  $> 13$ , Station 1/3 y-ratio  $< 1$ .

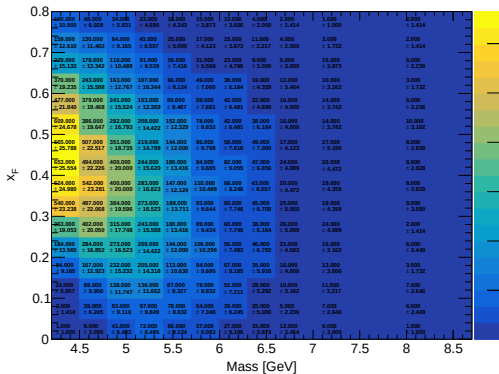
**Occupancy**  $D1, D2, D3 < 400$  hits each; Total  $D1 + D2 + D3 < 1000$ .

**Physics**  $Z_{\text{vertex}} \in [-320, -5]$ , Track Separation  $< 270$ ,  $|\cos \theta| < 0.5$ .



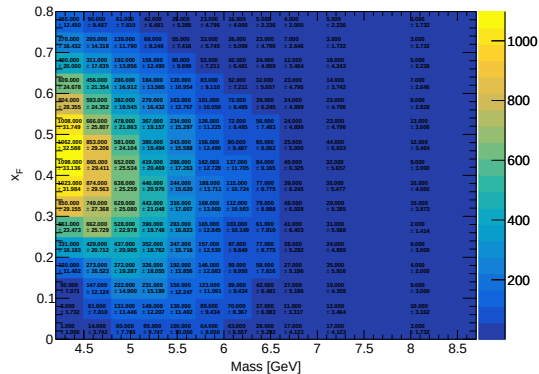
RS67

Total Yield (result) (LH2)



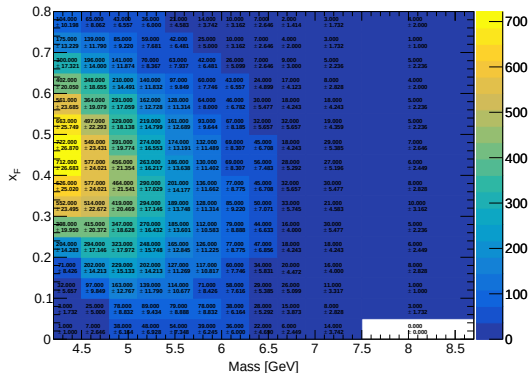
RS57-70

Total Yield (result) (LH2)



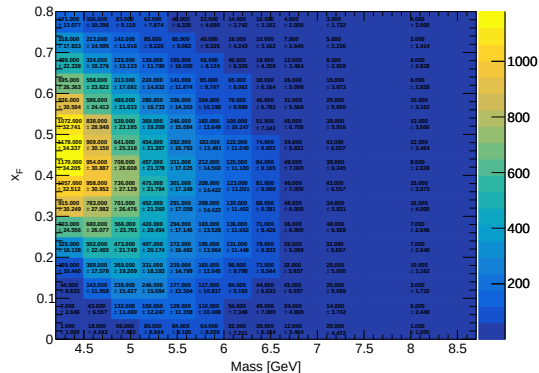
RS67

Total Yield (result) (LD2)



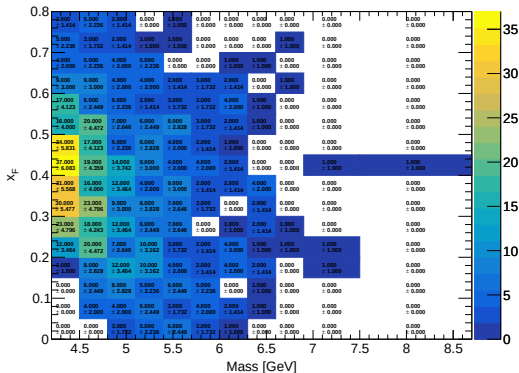
RS57-70

Total Yield (result) (LD2)



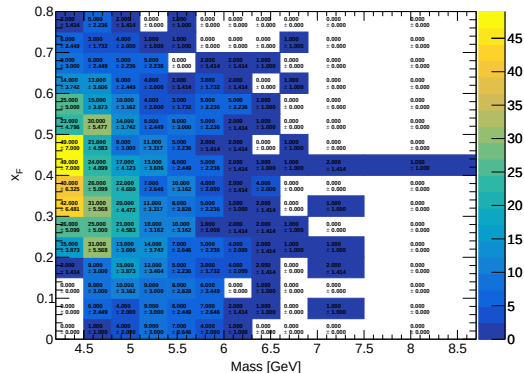
RS67

Total Yield (result) (Flask)



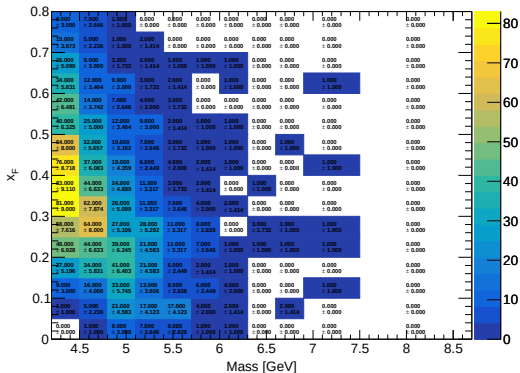
RS57-70

Total Yield (result) (Flask)



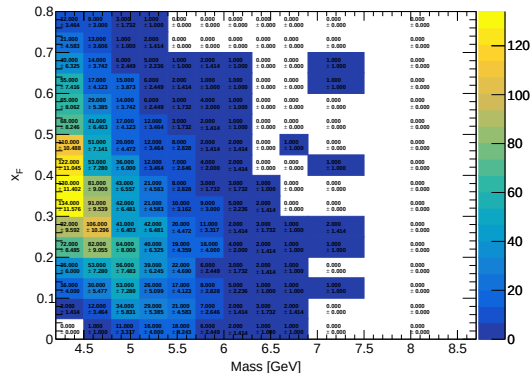
RS67

Mix Yield (result\_mix) (LH2)



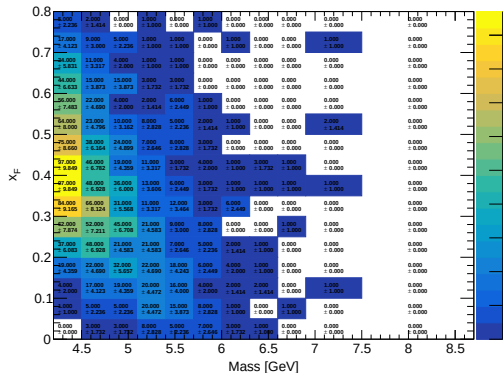
RS57-70

Mix Yield (result\_mix) (LH2)



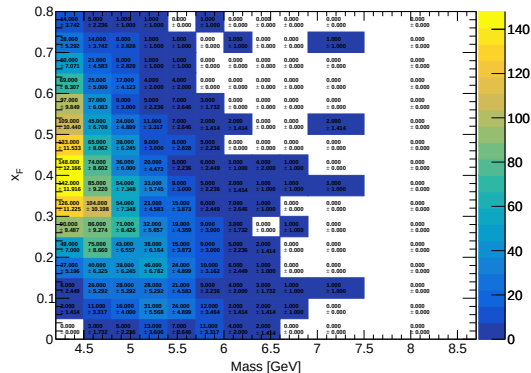
RS67

Mix Yield (result\_mix) (LD2)



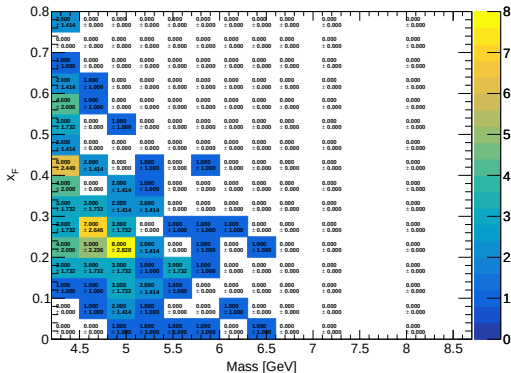
RS57-70

Mix Yield (result\_mix) (LD2)



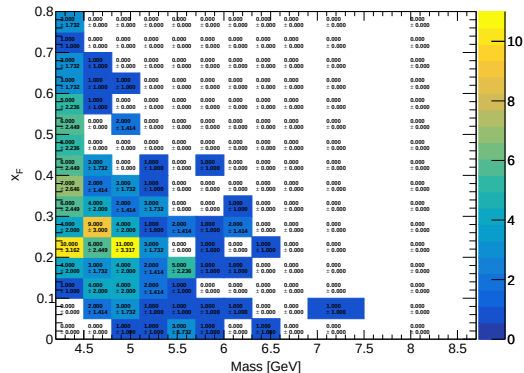
RS67

Mix Yield (result\_mix) (Flask)



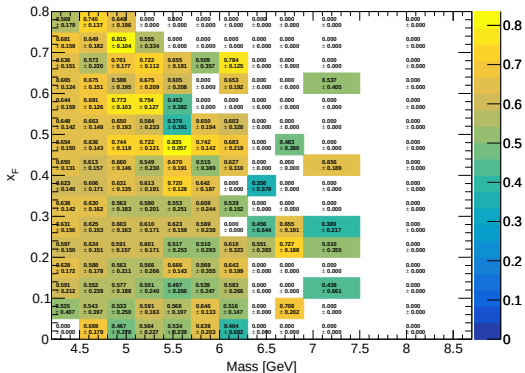
RS57-70

Mix Yield (result\_mix) (Flask)



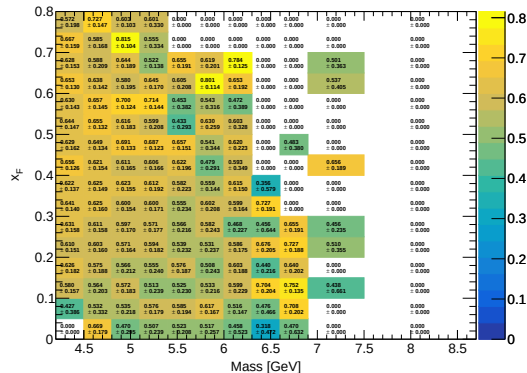
RS67

Avg Reco Eff (Mix) (LH2)



RS57-70

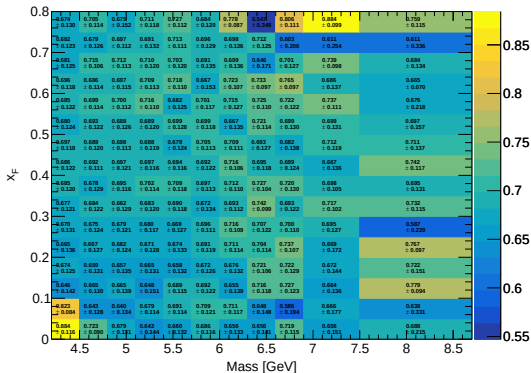
Avg Reco Eff (Mix) (LH2)



# E\_total\_reco\_LH2

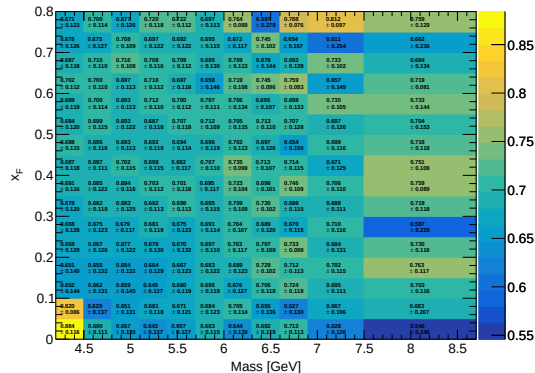
## RS67

Avg Reco Eff (Total) (LH2)



## RS57-70

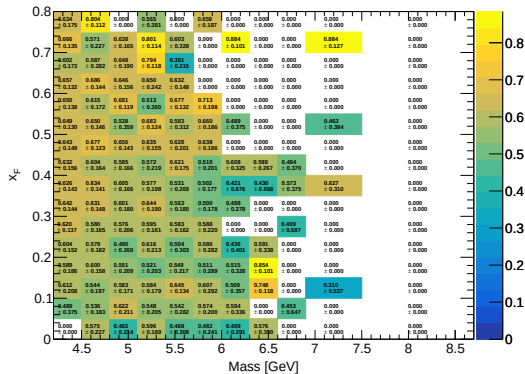
Avg Reco Eff (Total) (LH2)





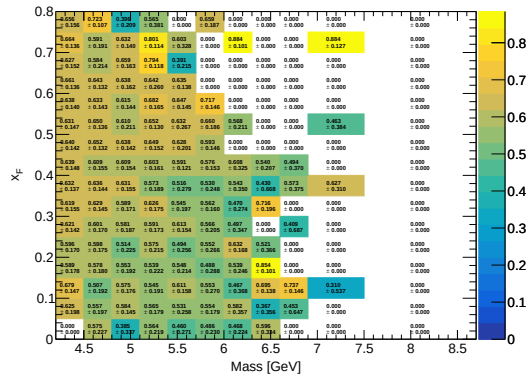
RS67

Avg Reco Eff (Mix) (LD2)



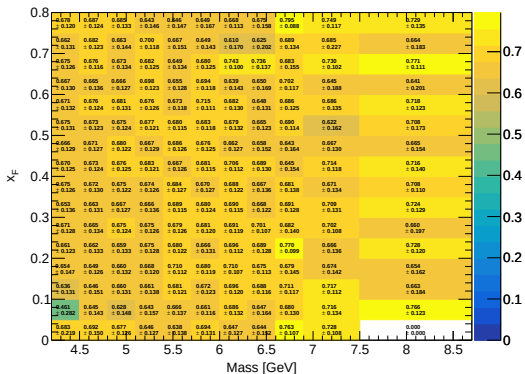
RS57-70

Avg Reco Eff (Mix) (LD2)



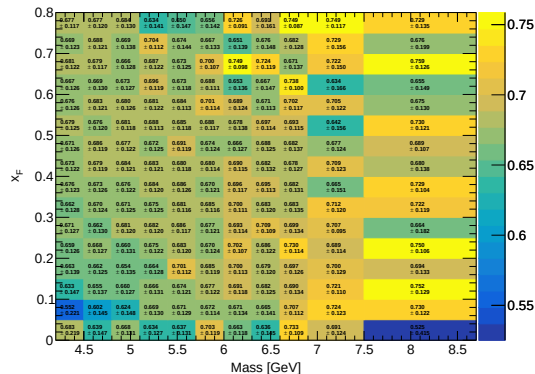
RS67

Avg Reco Eff (Total) (LD2)



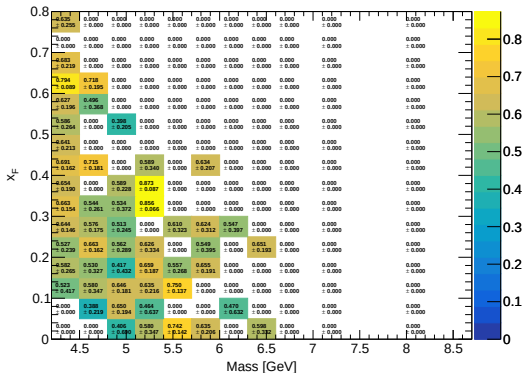
RS57-70

Avg Reco Eff (Total) (LD2)



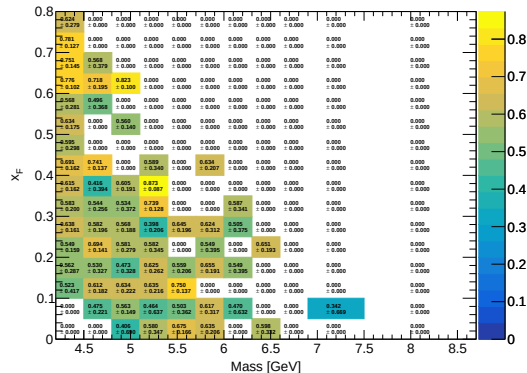
RS67

Avg Reco Eff (Mix) (Flask)



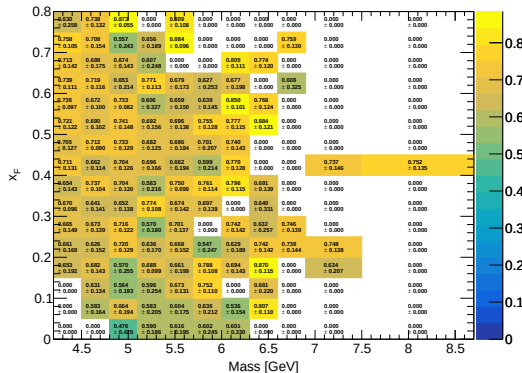
RS57-70

Avg Reco Eff (Mix) (Flask)



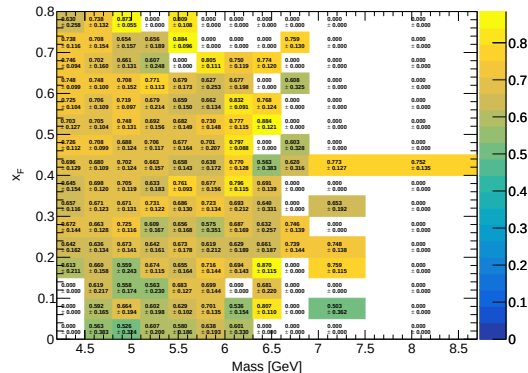
## RS67

Avg Reco Eff (Total) (Flask)



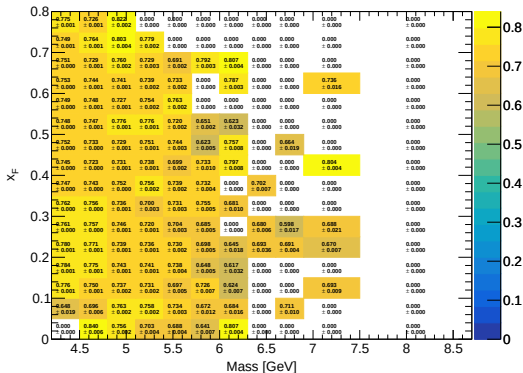
## RS57-70

Avg Reco Eff (Total) (Flask)



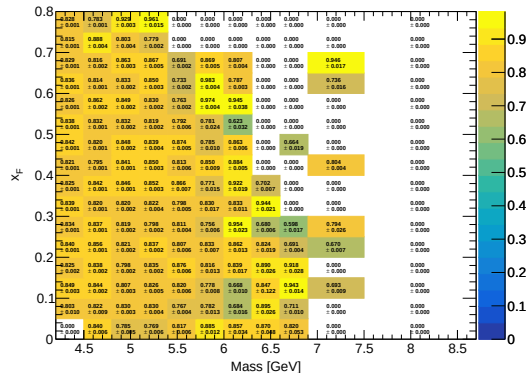
RS67

Avg Hodo Eff (Mix) (LH2)



RS57-70

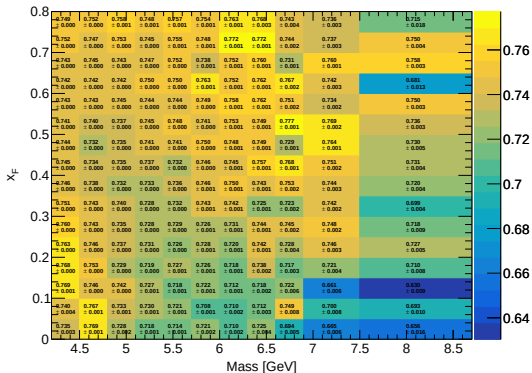
Avg Hodo Eff (Mix) (LH2)



# E\_total\_hodo\_LH2

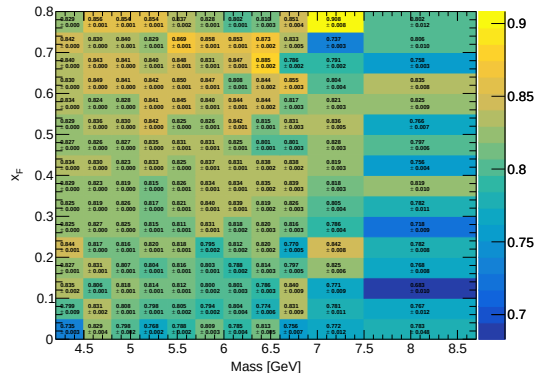
## RS67

Avg Hodo Eff (Total) (LH2)



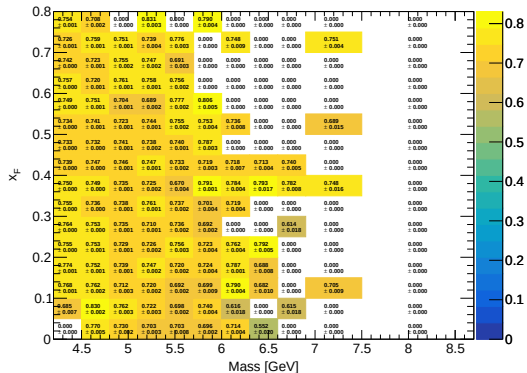
## RS57-70

Avg Hodo Eff (Total) (LH2)



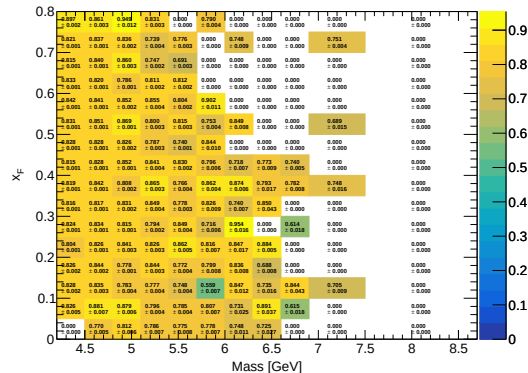
RS67

Avg Hodo Eff (Mix) (LD2)



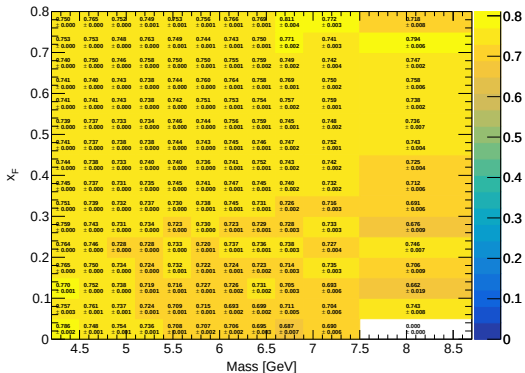
RS57-70

Avg Hodo Eff (Mix) (LD2)



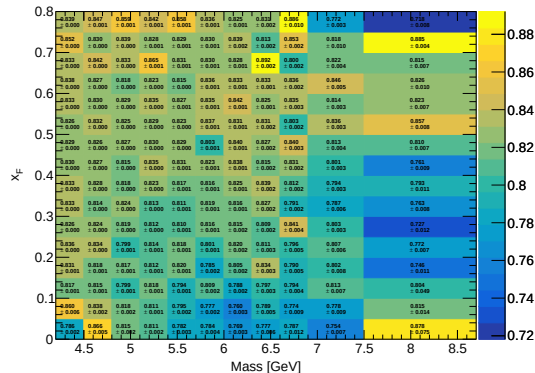
## RS67

Avg Hodo Eff (Total) (LD2)

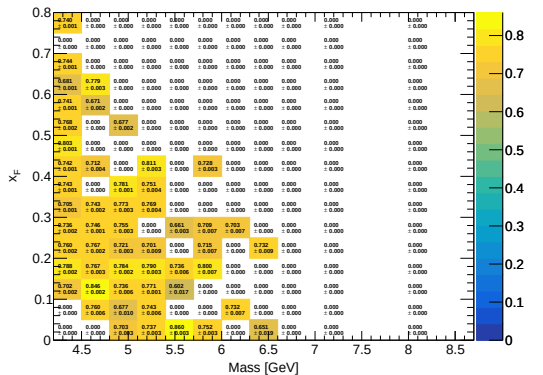


## RS57-70

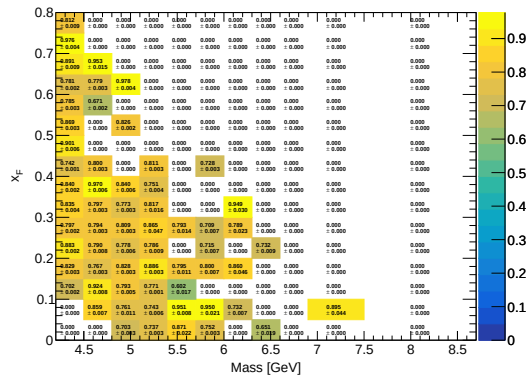
Avg Hodo Eff (Total) (LD2)







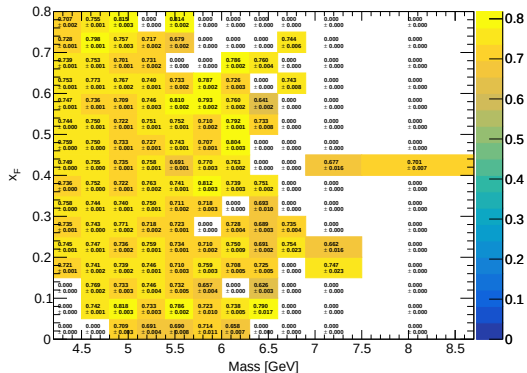
**RS57-70**  
Avg Hodo Eff (Mix) (Flask)



# E\_total\_hodo\_Flask

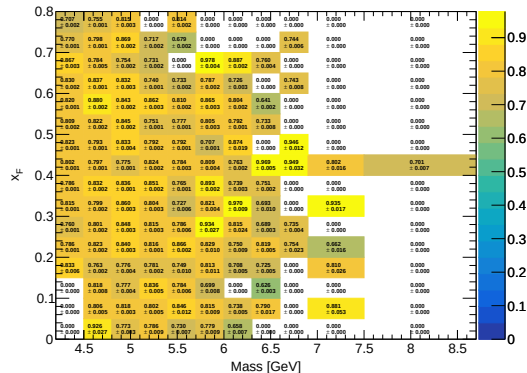
## RS67

Avg Hodo Eff (Total) (Flask)



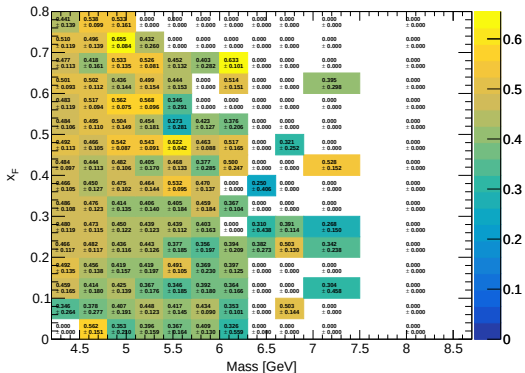
## RS57-70

Avg Hodo Eff (Total) (Flask)



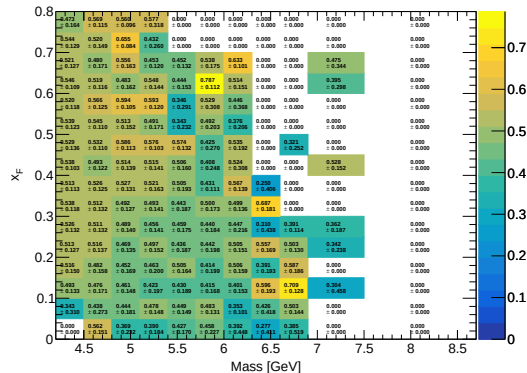
RS67

Avg Final Eff (Mix) (LH2)



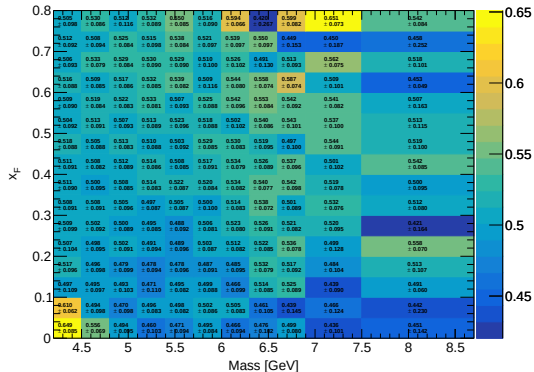
RS57-70

Avg Final Eff (Mix) (LH2)



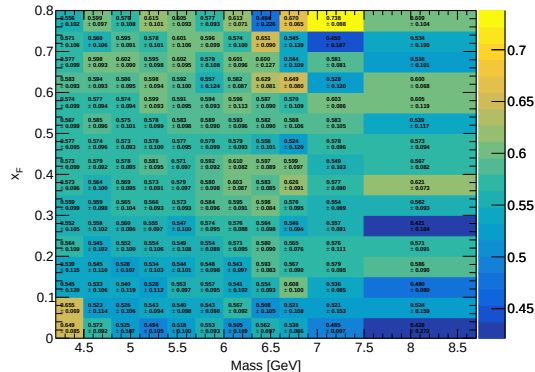
RS67

Avg Final Eff (Total) (LH2)



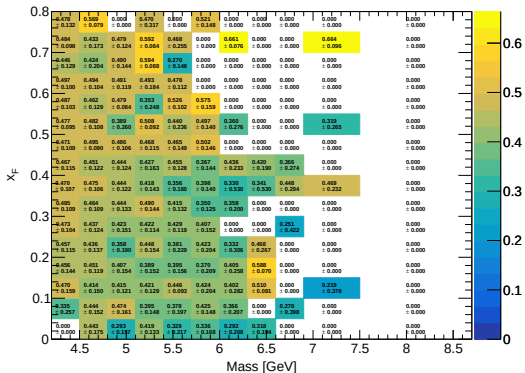
RS57-70

Avg Final Eff (Total) (LH2)



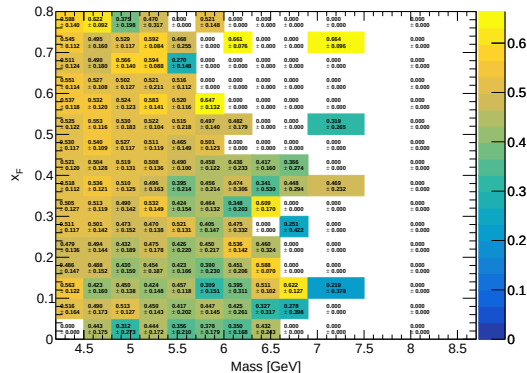
RS67

Avg Final Eff (Mix) (LD2)



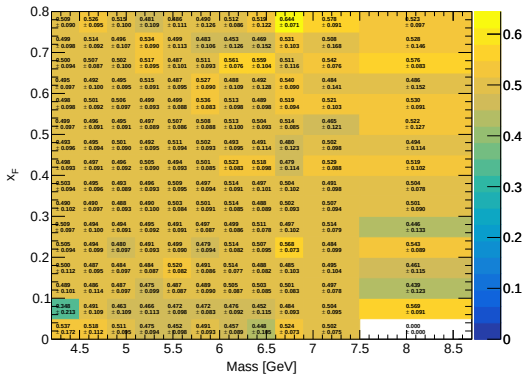
RS57-70

Avg Final Eff (Mix) (LD2)



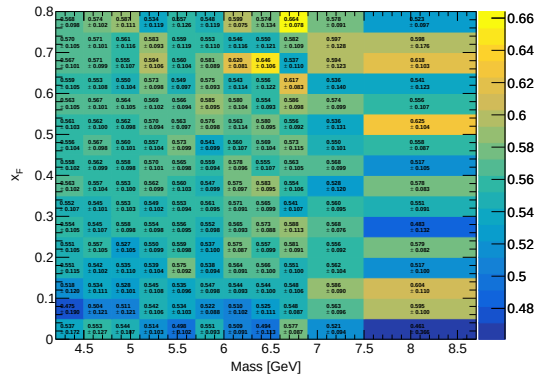
## RS67

Avg Final Eff (Total) (LD2)



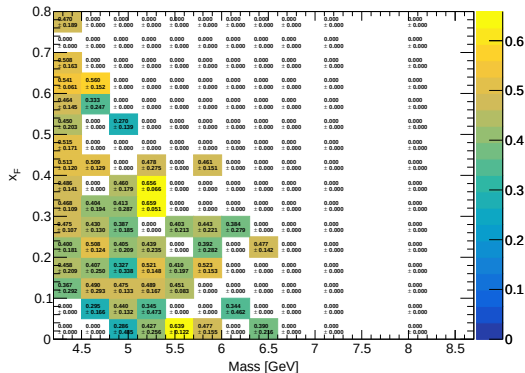
## RS57-70

Avg Final Eff (Total) (LD2)



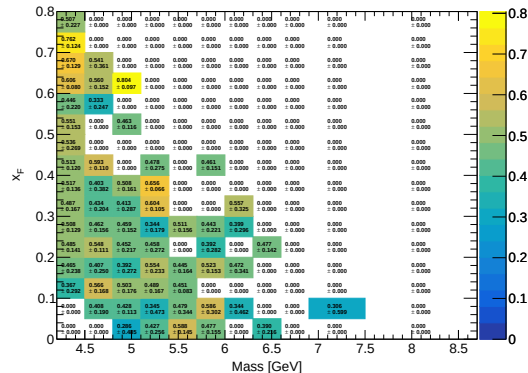
RS67

Avg Final Eff (Mix) (Flask)



RS57-70

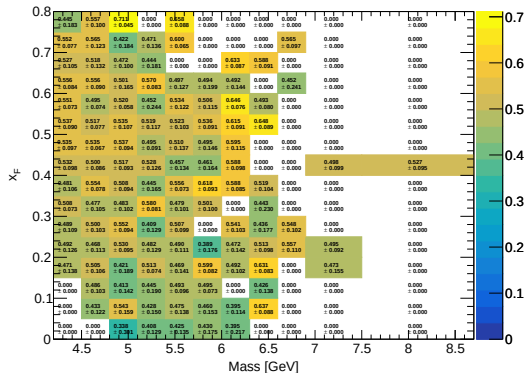
Avg Final Eff (Mix) (Flask)



# E\_total\_final\_Flask

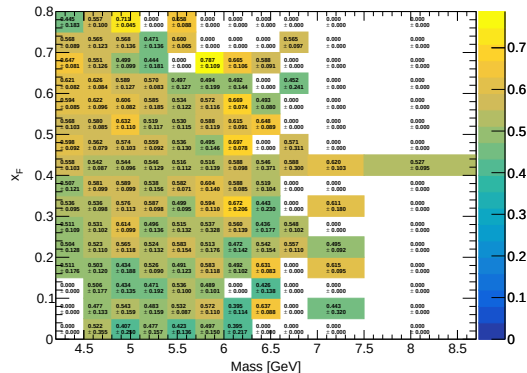
## RS67

Avg Final Eff (Total) (Flask)



## RS57-70

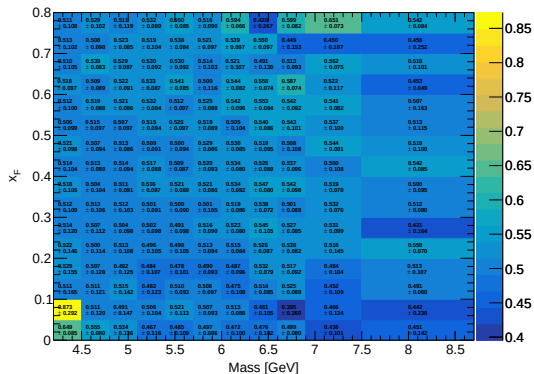
Avg Final Eff (Total) (Flask)





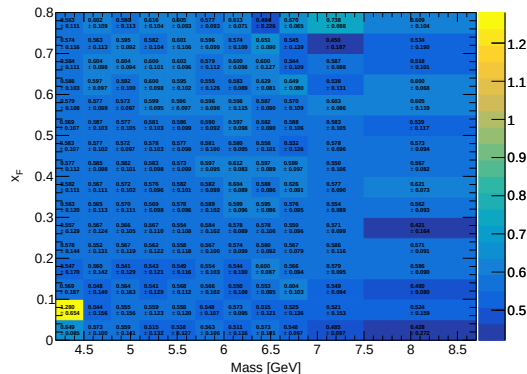
## RS67

Avg Signal Efficiency (LH2)

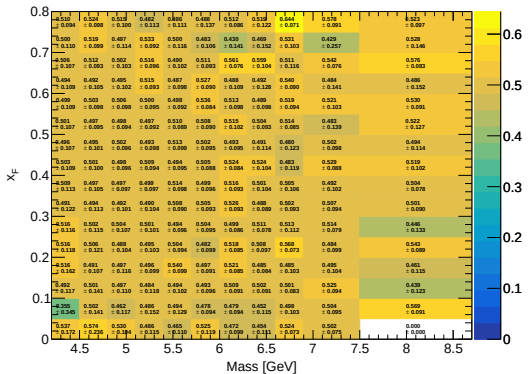


## RS57-70

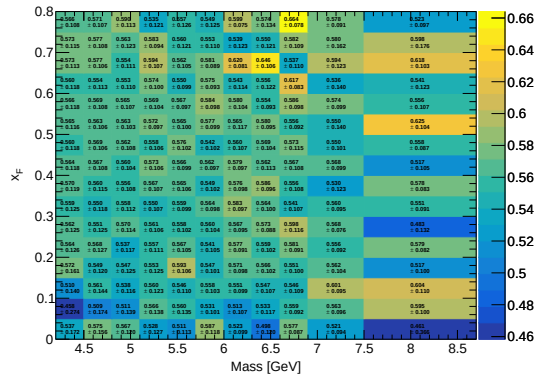
Avg Signal Efficiency (LH2)



**RS67**  
Avg Signal Efficiency (LD2)

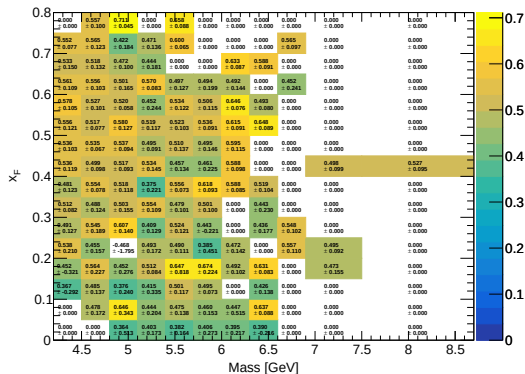


**RS57-70**  
Avg Signal Efficiency (LD2)



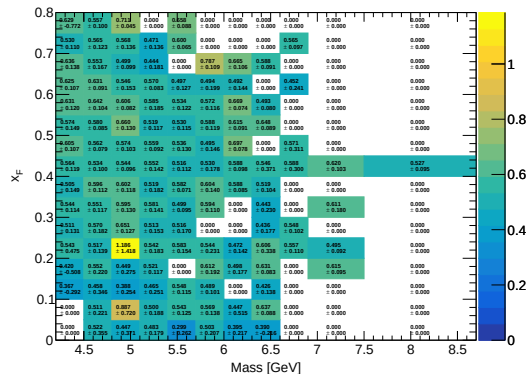
## RS67

Avg Signal Efficiency (Flask)



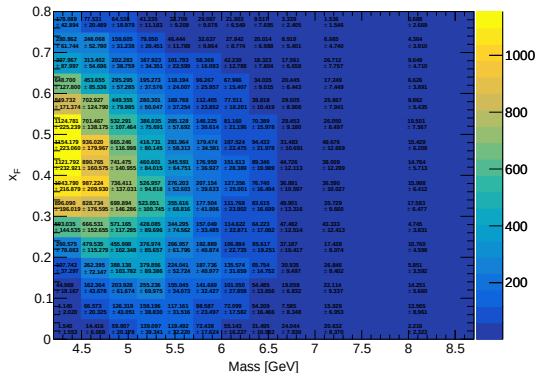
## RS57-70

Avg Signal Efficiency (Flask)



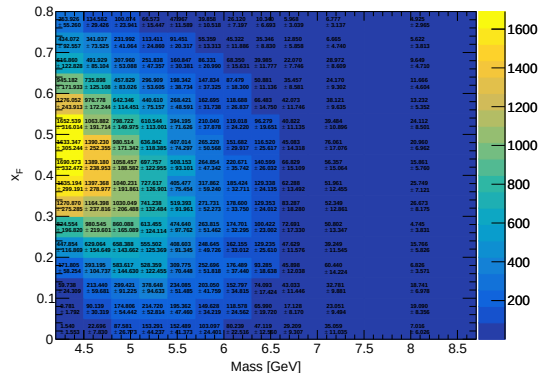
RS67

Corrected Yield (Total Error) (LH2)



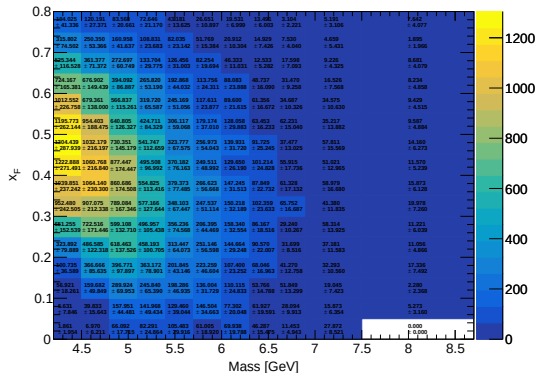
RS57-70

Corrected Yield (Total Error) (LH2)



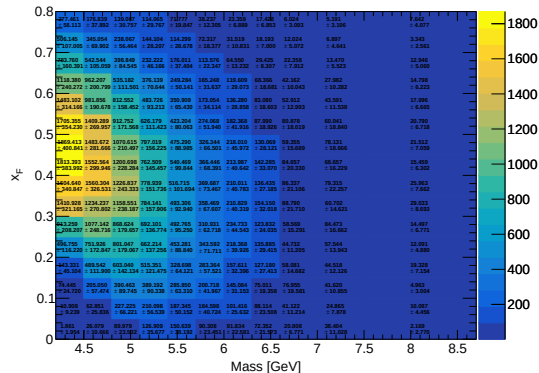
RS67

Corrected Yield (Total Error) (LD2)



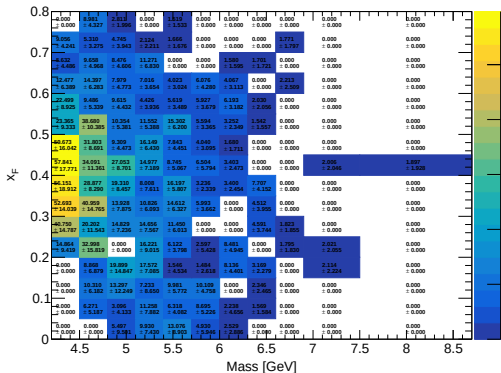
RS57-70

Corrected Yield (Total Error) (LD2)



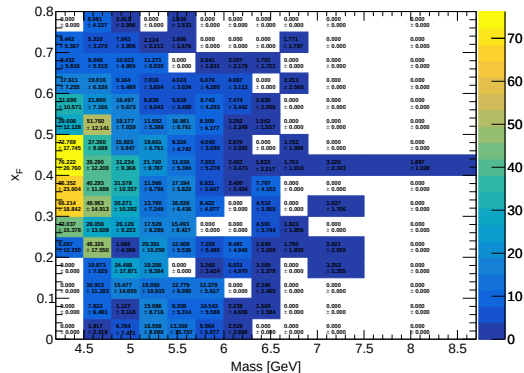
RS67

Corrected Yield (Total Error) (Flask)



RS57-70

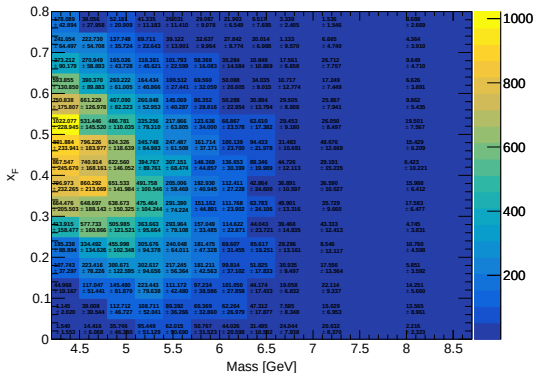
Corrected Yield (Total Error) (Flask)



# Y\_corrected\_Subtracted\_LH2

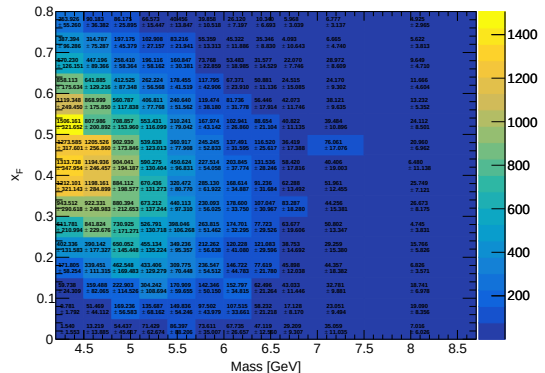
## RS67

Corrected Yield (LH2 - Flask)



## RS57-70

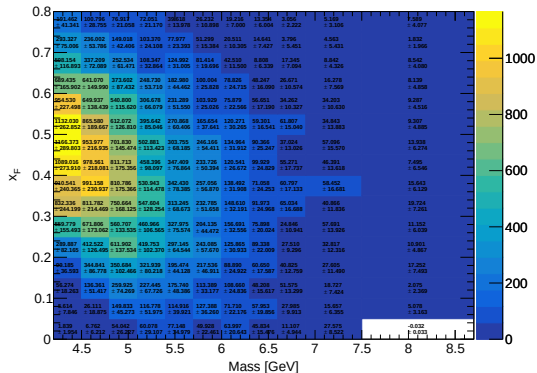
Corrected Yield (LH2 - Flask)



# Y\_corrected\_Subtracted\_LD2

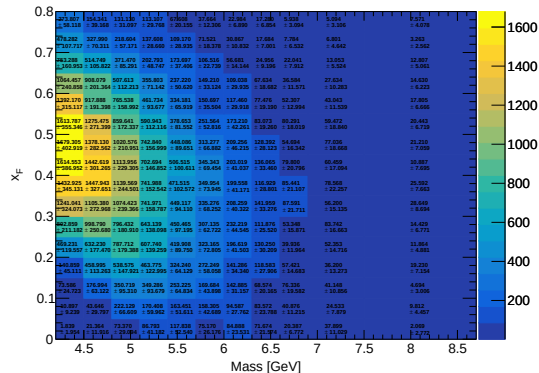
## RS67

Corrected Yield (LD2 - LH2 - Flask)



## RS57-70

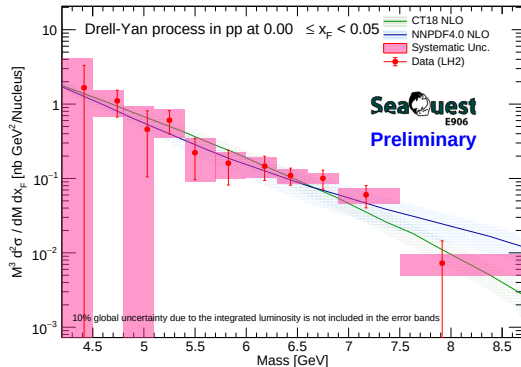
Corrected Yield (LD2 - LH2 - Flask)



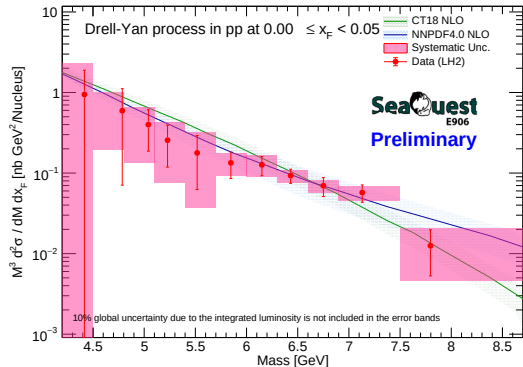


# CrossSection LH2 $0.00 \leq x_F < 0.05$ (Centroid)

## RS67

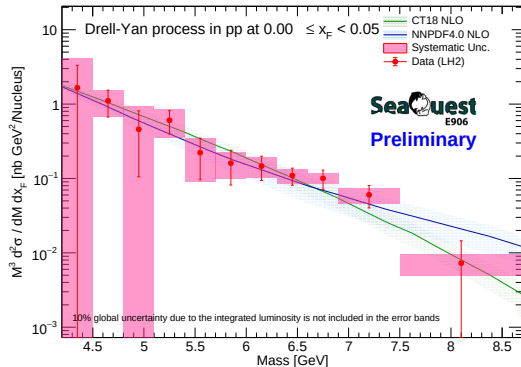


## RS57-70

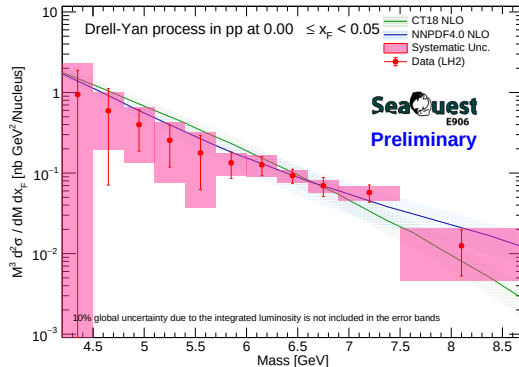


# CrossSection LH2 $0.00 \leq x_F < 0.05$ (GeoCenter)

## RS67

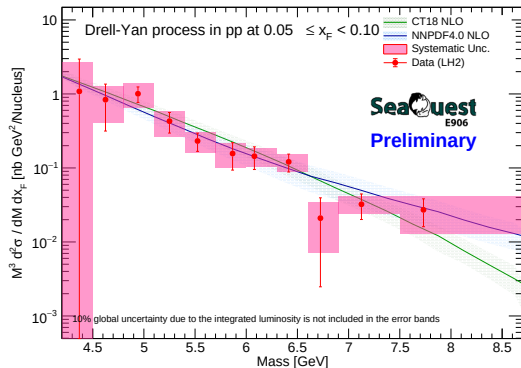


## RS57-70

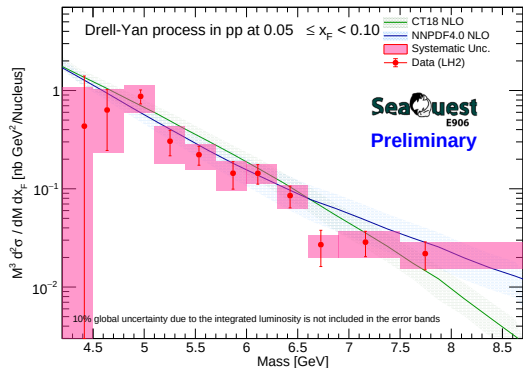


# CrossSection LH2 $0.05 \leq x_F < 0.10$ (Centroid)

## RS67

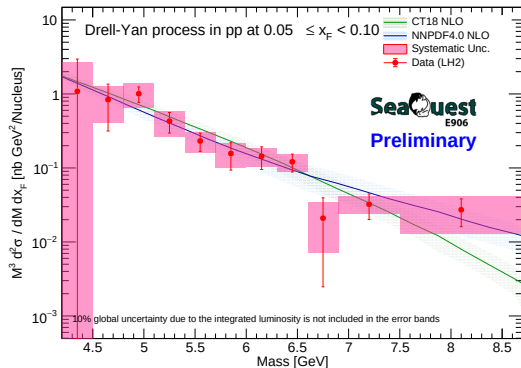


## RS57-70

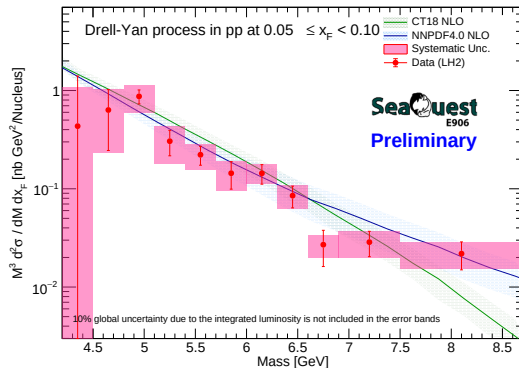


# CrossSection LH2 $0.05 \leq x_F < 0.10$ (GeoCenter)

## RS67

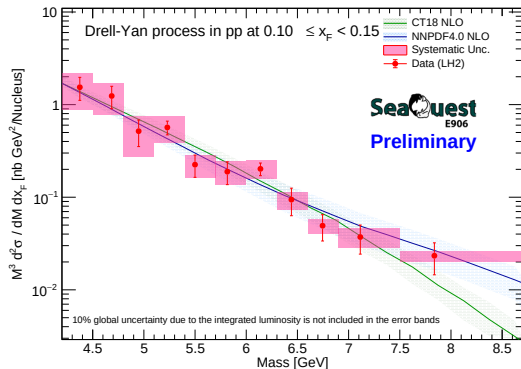


## RS57-70

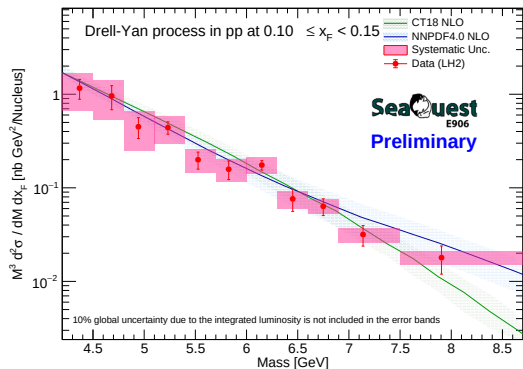


# CrossSection LH2 $0.10 \leq x_F < 0.15$ (Centroid)

## RS67

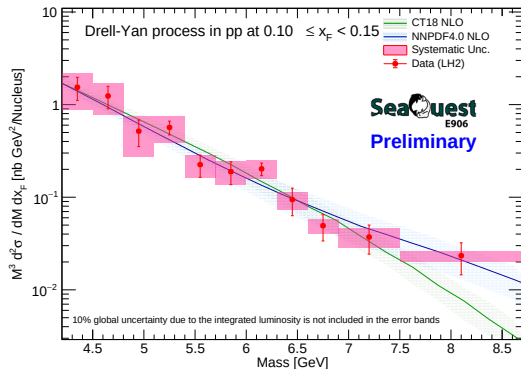


## RS57-70

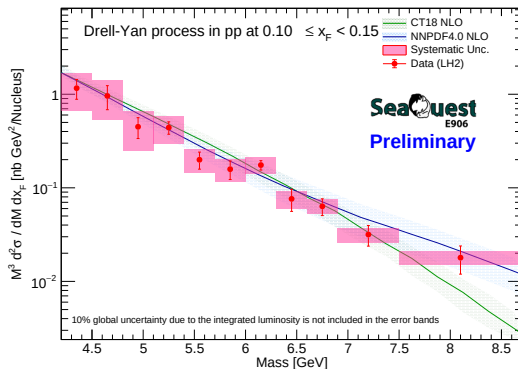


# CrossSection LH2 $0.10 \leq x_F < 0.15$ (GeoCenter)

## RS67

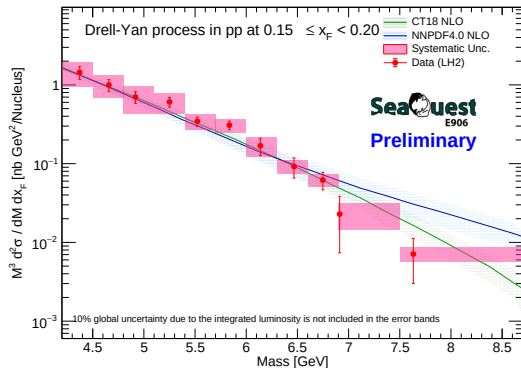


## RS57-70

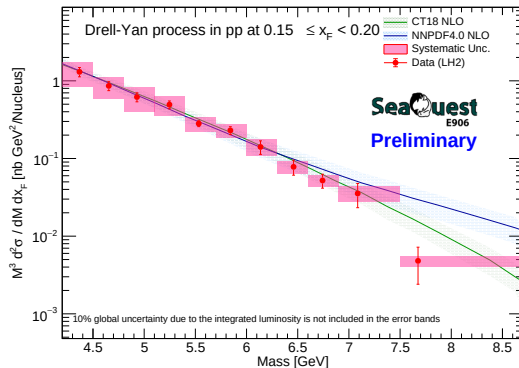


# CrossSection LH2 $0.15 \leq x_F < 0.20$ (Centroid)

## RS67

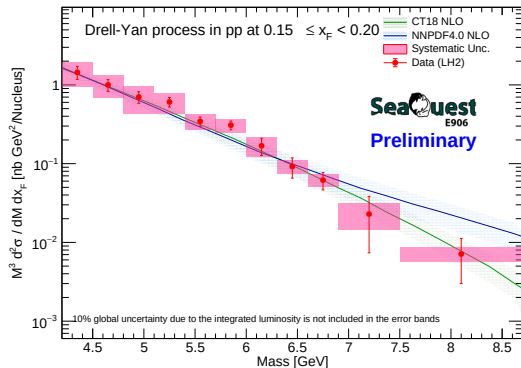


## RS57-70

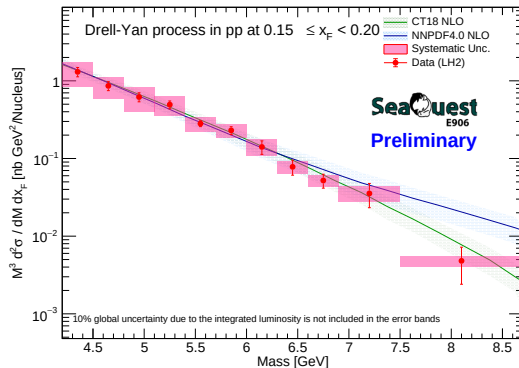


# CrossSection LH2 $0.15 \leq x_F < 0.20$ (GeoCenter)

## RS67



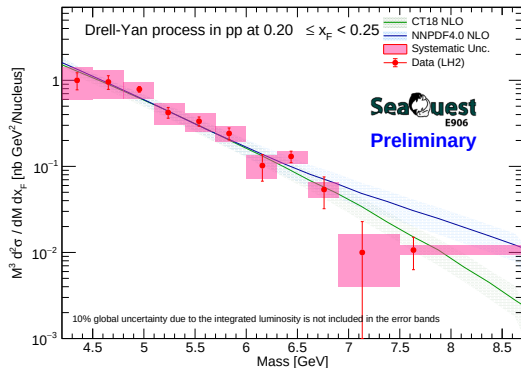
## RS57-70



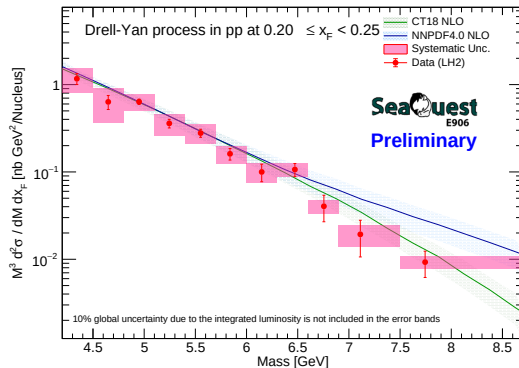


# CrossSection LH2 $0.20 \leq x_F < 0.25$ (Centroid)

## RS67

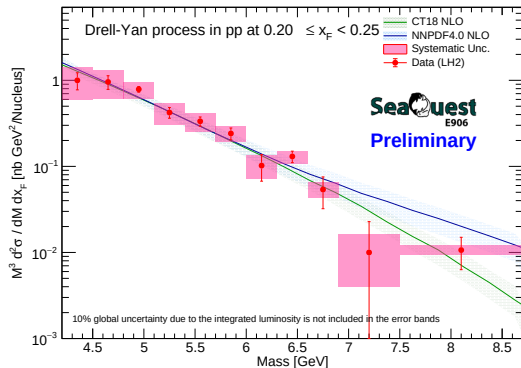


## RS57-70

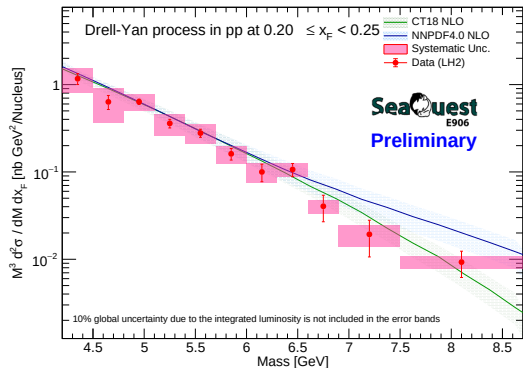


# CrossSection LH2 $0.20 \leq x_F < 0.25$ (GeoCenter)

## RS67

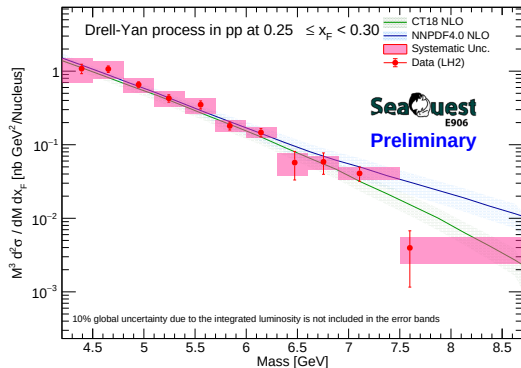


## RS57-70

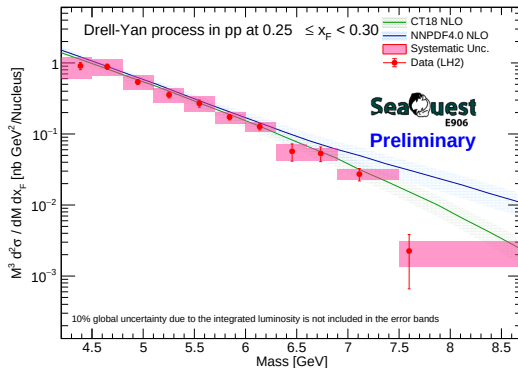


# CrossSection LH2 $0.25 \leq x_F < 0.30$ (Centroid)

## RS67

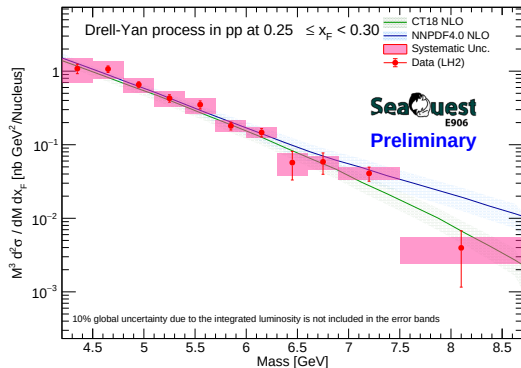


## RS57-70

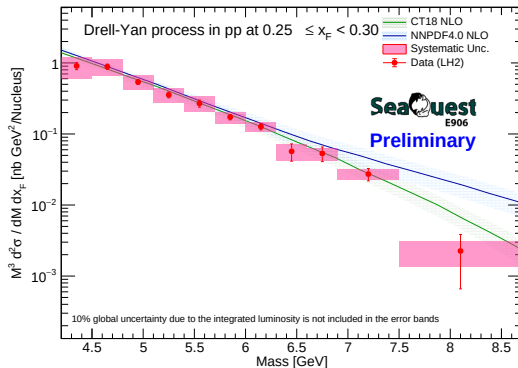


# CrossSection LH2 $0.25 \leq x_F < 0.30$ (GeoCenter)

## RS67

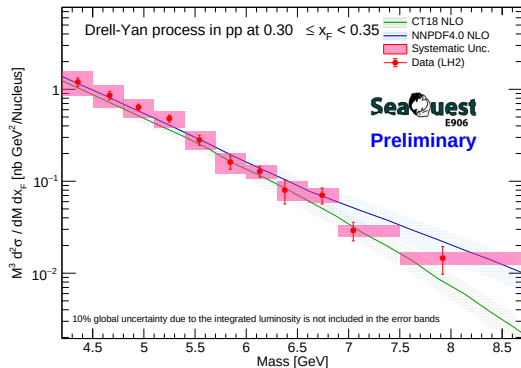


## RS57-70

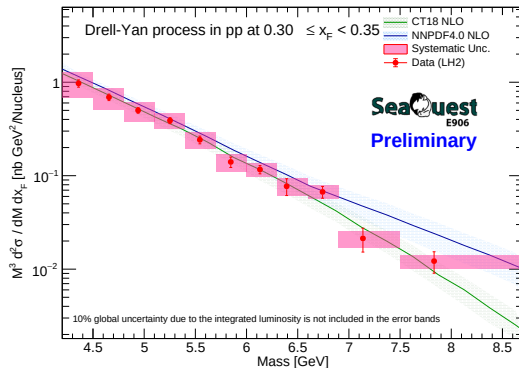


# CrossSection LH2 $0.30 \leq x_F < 0.35$ (Centroid)

## RS67

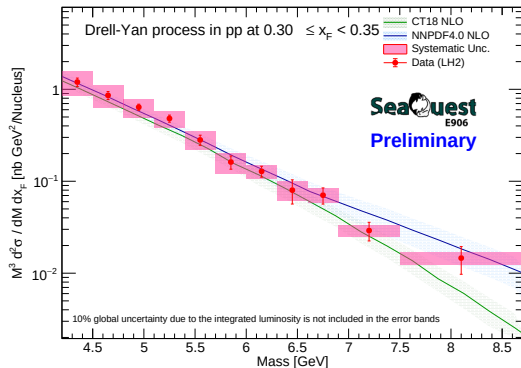


## RS57-70

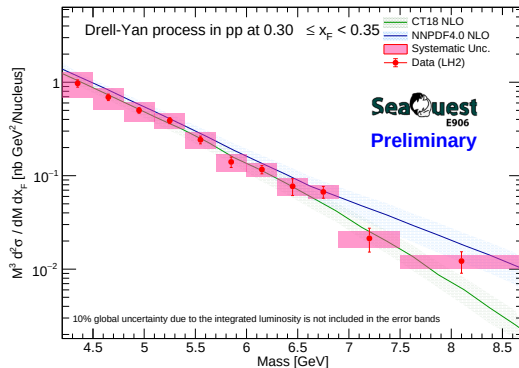


# CrossSection LH2 $0.30 \leq x_F < 0.35$ (GeoCenter)

## RS67

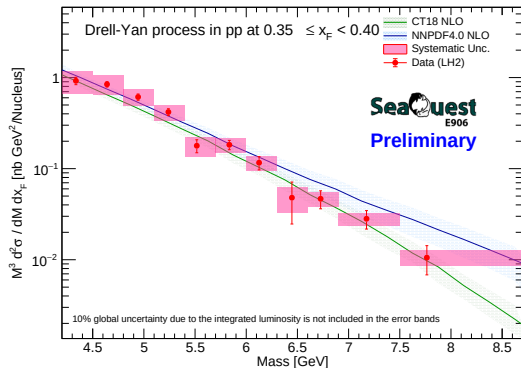


## RS57-70

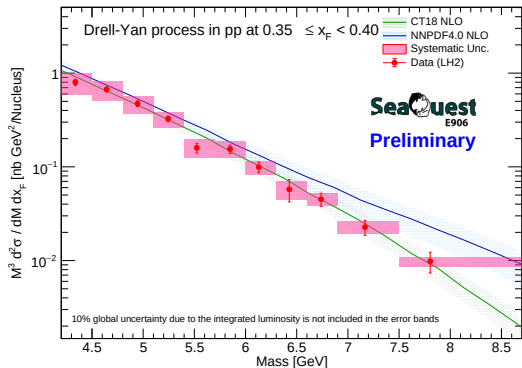


# CrossSection LH2 $0.35 \leq x_F < 0.40$ (Centroid)

## RS67

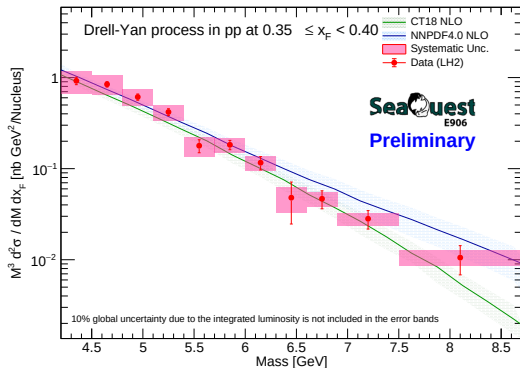


## RS57-70

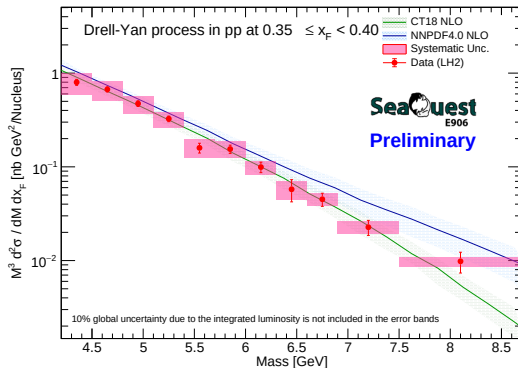


# CrossSection LH2 $0.35 \leq x_F < 0.40$ (GeoCenter)

## RS67



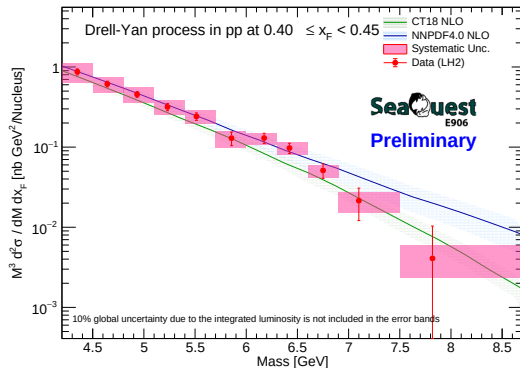
## RS57-70



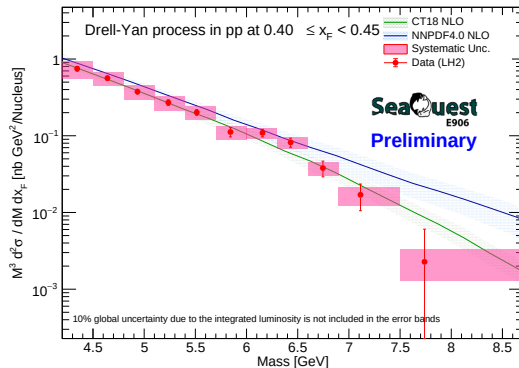


# CrossSection LH2 $0.40 \leq x_F < 0.45$ (Centroid)

## RS67

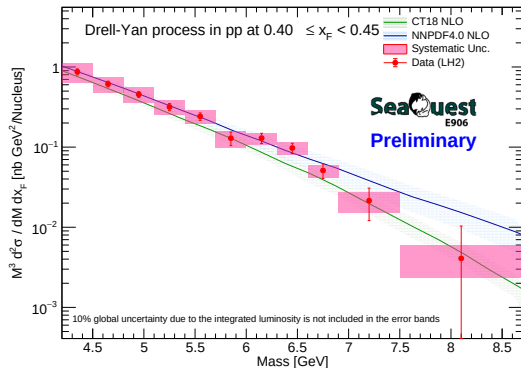


## RS57-70

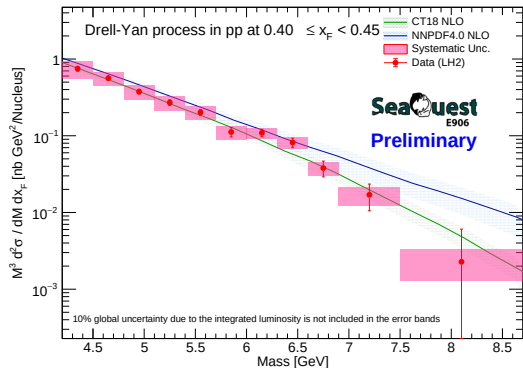


# CrossSection LH2 $0.40 \leq x_F < 0.45$ (GeoCenter)

## RS67

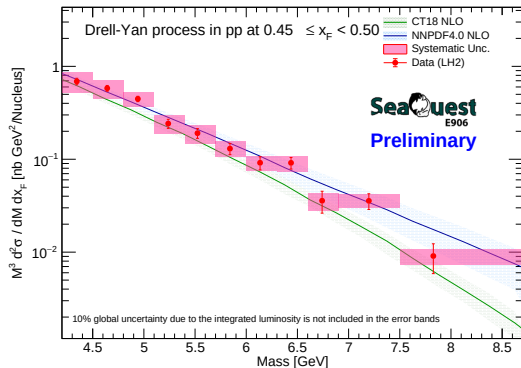


## RS57-70

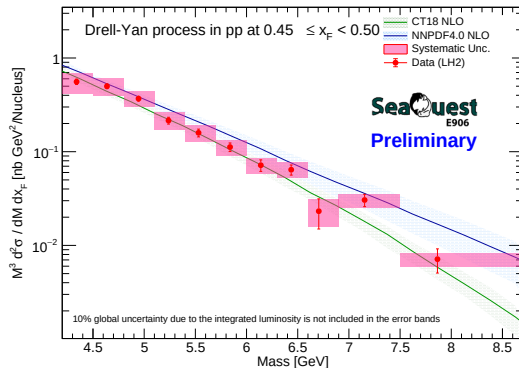


# CrossSection LH2 $0.45 \leq x_F < 0.50$ (Centroid)

## RS67

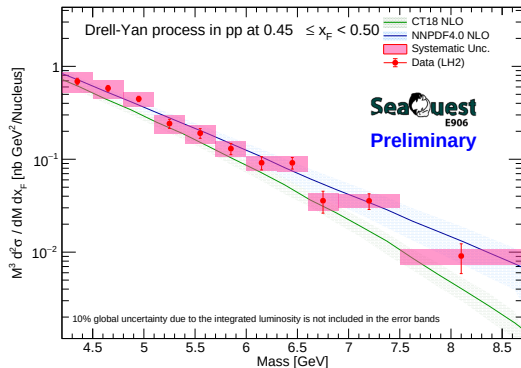


## RS57-70

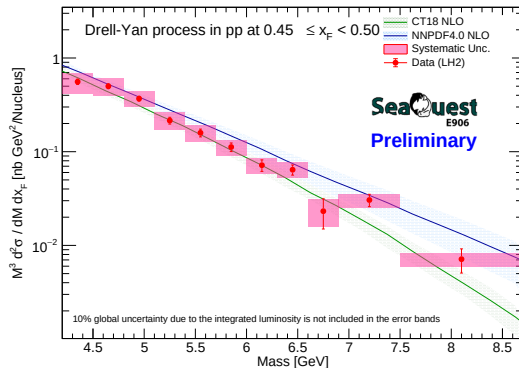


# CrossSection LH2 $0.45 \leq x_F < 0.50$ (GeoCenter)

## RS67

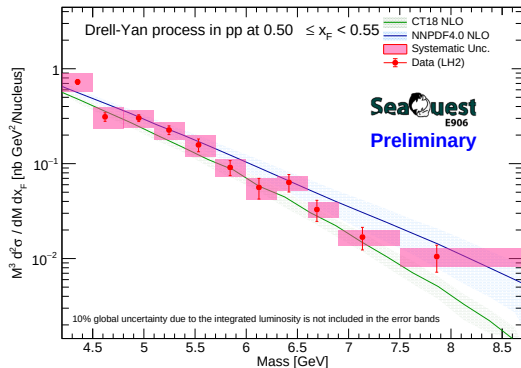


## RS57-70

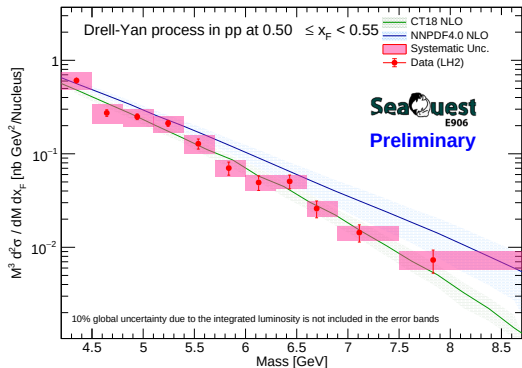


# CrossSection LH2 $0.50 \leq x_F < 0.55$ (Centroid)

## RS67

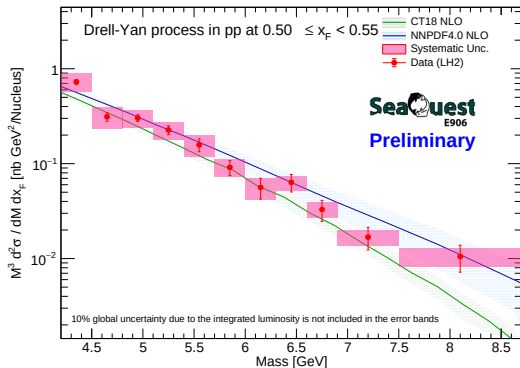


## RS57-70

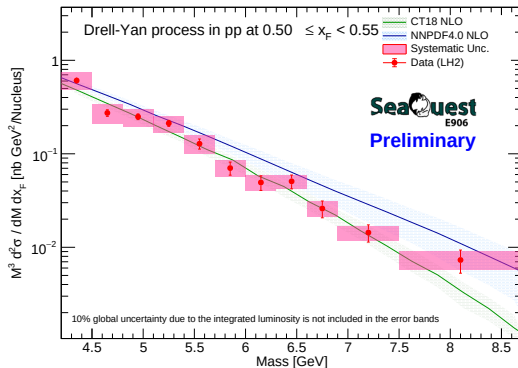


# CrossSection LH2 $0.50 \leq x_F < 0.55$ (GeoCenter)

## RS67

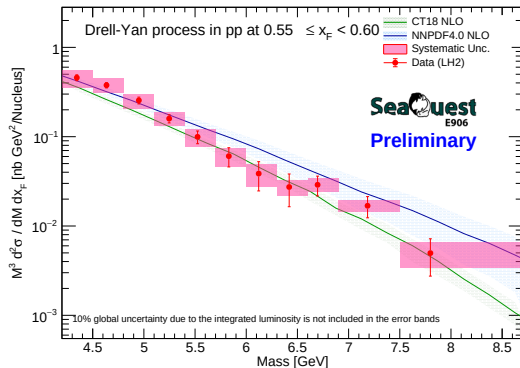


## RS57-70

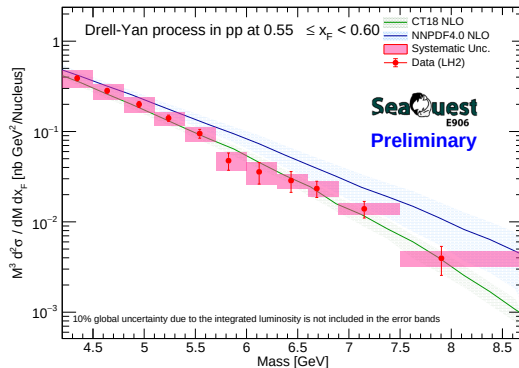


# CrossSection LH2 $0.55 \leq x_F < 0.60$ (Centroid)

## RS67

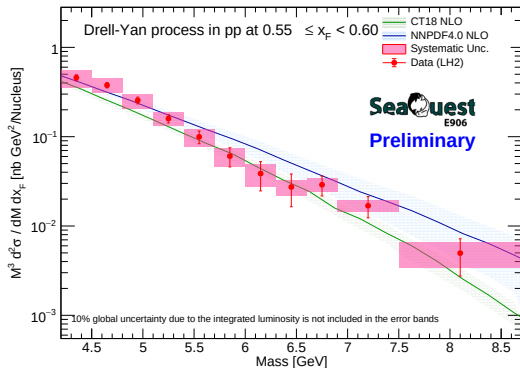


## RS57-70

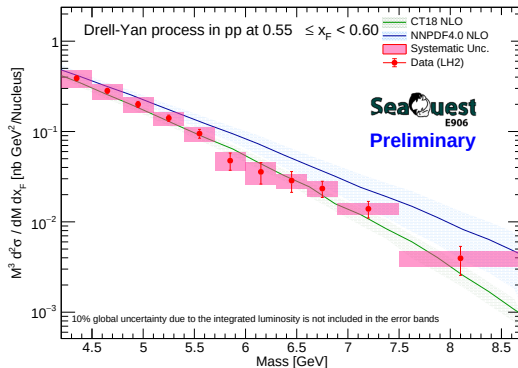


# CrossSection LH2 $0.55 \leq x_F < 0.60$ (GeoCenter)

## RS67



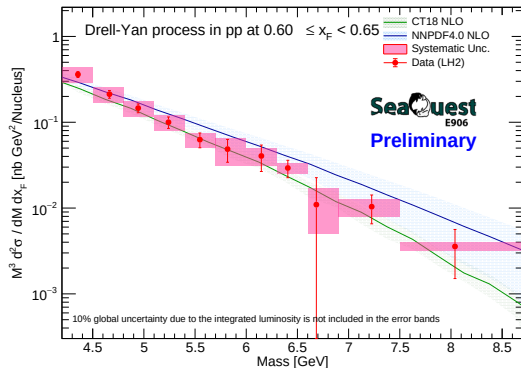
## RS57-70



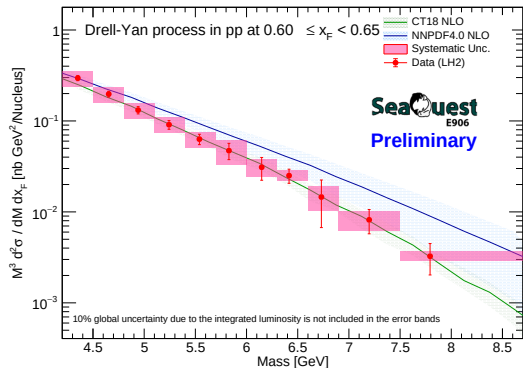


# CrossSection LH2 $0.60 \leq x_F < 0.65$ (Centroid)

## RS67

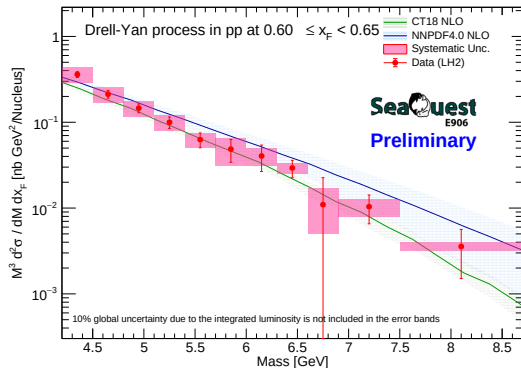


## RS57-70

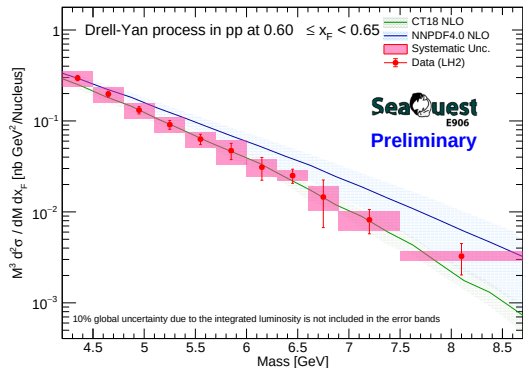


# CrossSection LH2 $0.60 \leq x_F < 0.65$ (GeoCenter)

## RS67

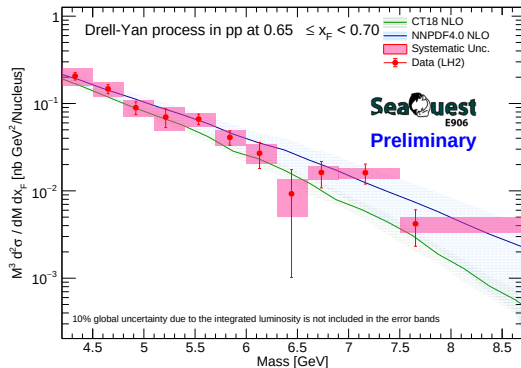


## RS57-70

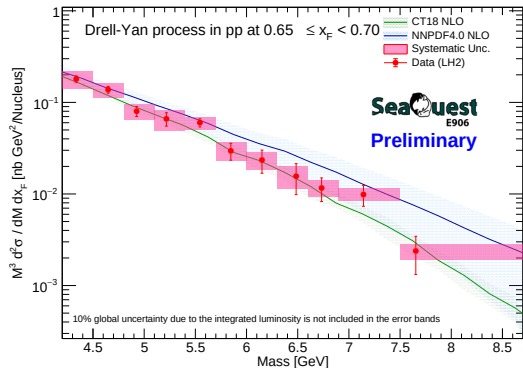


# CrossSection LH2 $0.65 \leq x_F < 0.70$ (Centroid)

## RS67

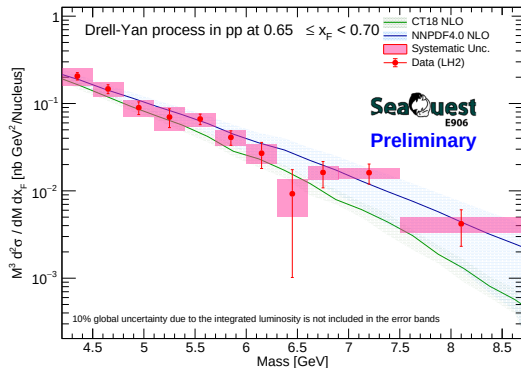


## RS57-70

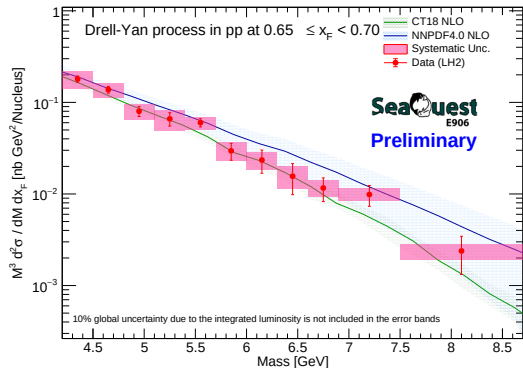


# CrossSection LH2 $0.65 \leq x_F < 0.70$ (GeoCenter)

## RS67

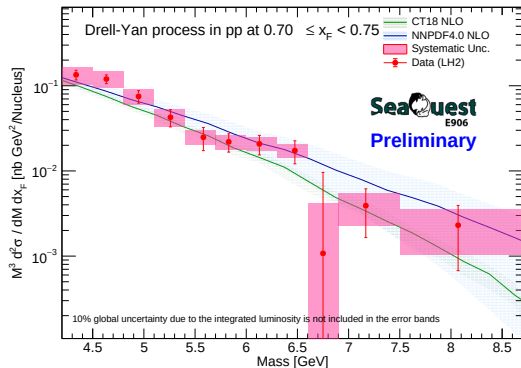


## RS57-70

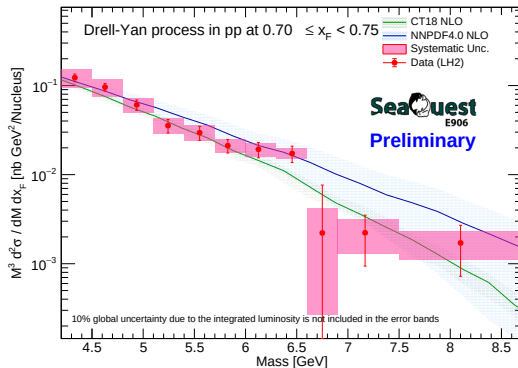


# CrossSection LH2 $0.70 \leq x_F < 0.75$ (Centroid)

## RS67

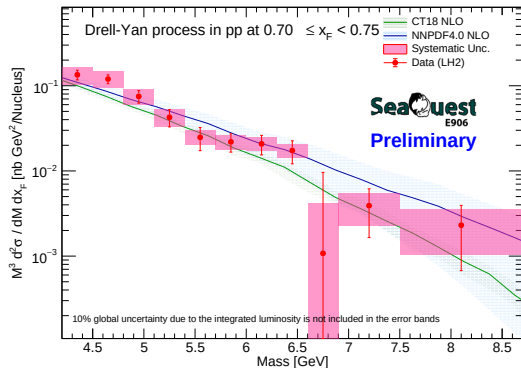


## RS57-70

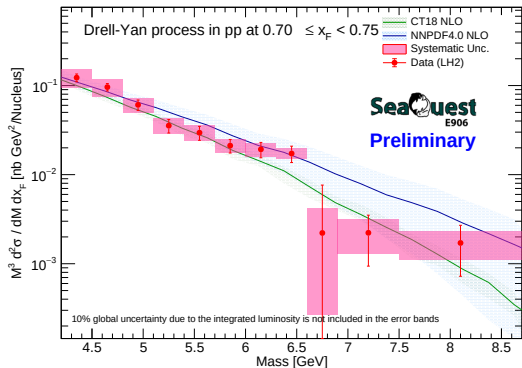


# CrossSection LH2 $0.70 \leq x_F < 0.75$ (GeoCenter)

## RS67

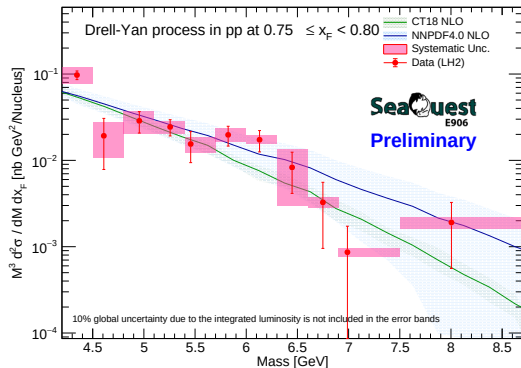


## RS57-70

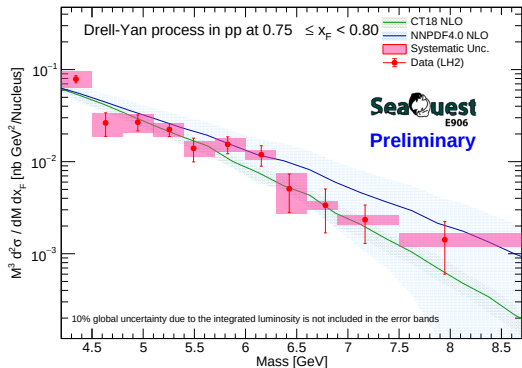


# CrossSection LH2 $0.75 \leq x_F < 0.80$ (Centroid)

## RS67

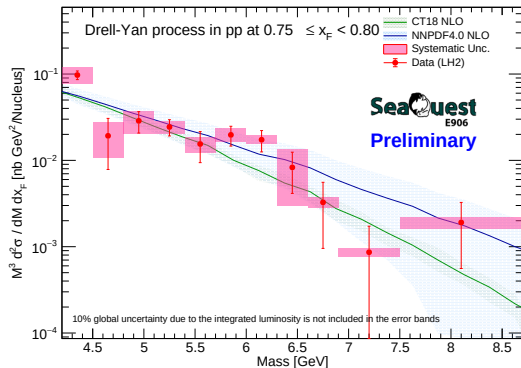


## RS57-70

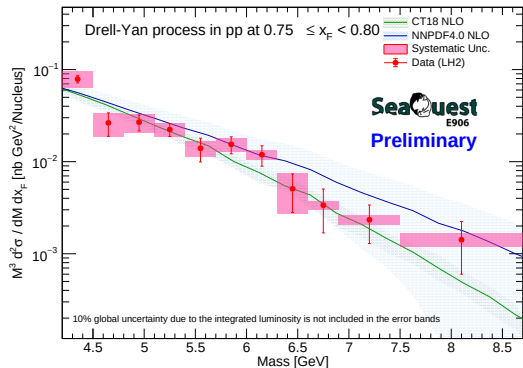


# CrossSection LH2 $0.75 \leq x_F < 0.80$ (GeoCenter)

## RS67



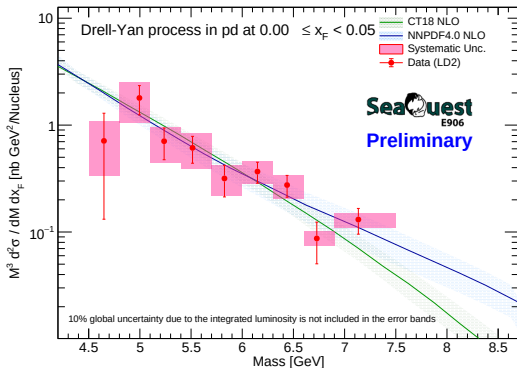
## RS57-70



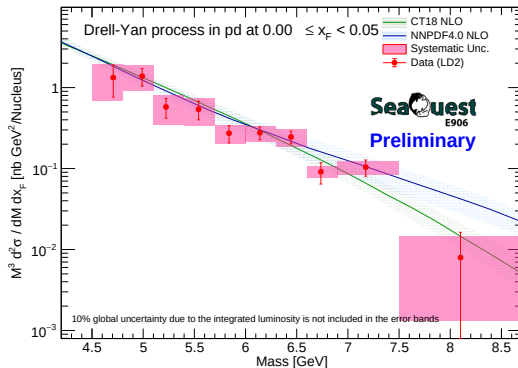


# CrossSection LD2 $0.00 \leq x_F < 0.05$ (Centroid)

## RS67

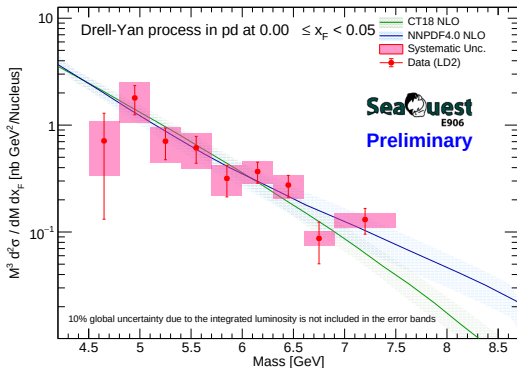


## RS57-70

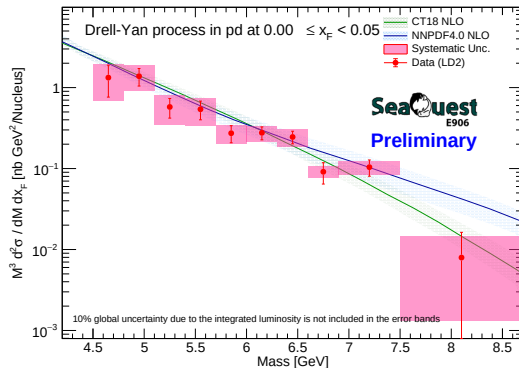


# CrossSection LD2 $0.00 \leq x_F < 0.05$ (GeoCenter)

## RS67

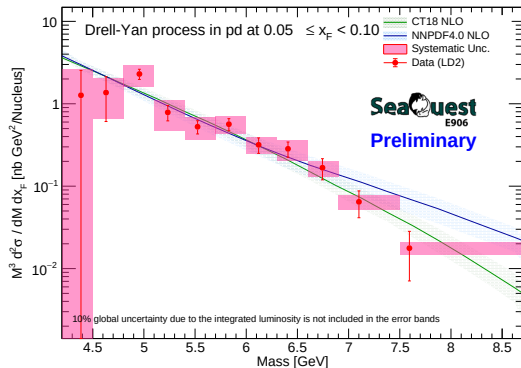


## RS57-70

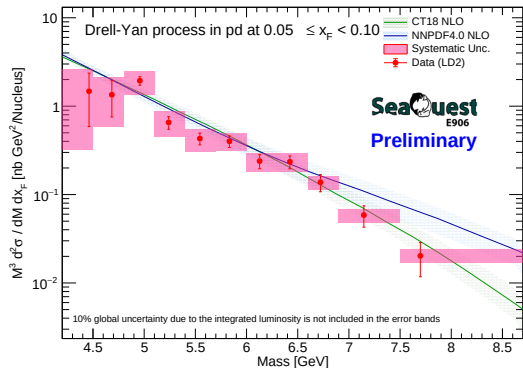


# CrossSection LD2 $0.05 \leq x_F < 0.10$ (Centroid)

## RS67

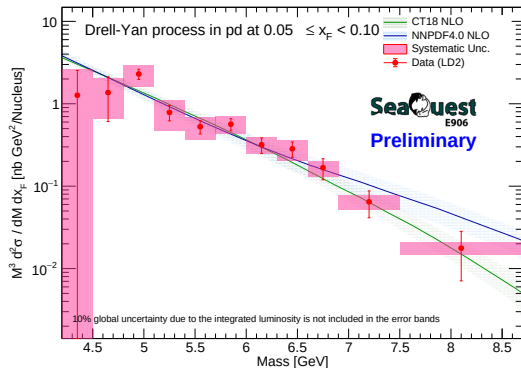


## RS57-70

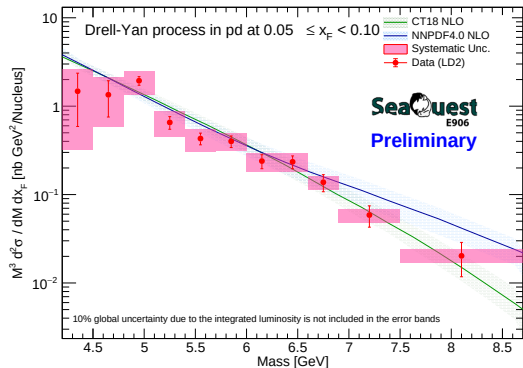


# CrossSection LD2 $0.05 \leq x_F < 0.10$ (GeoCenter)

## RS67

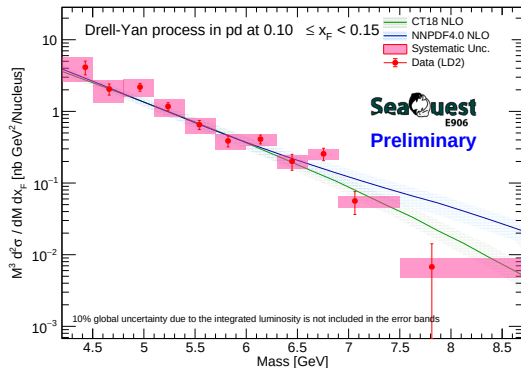


## RS57-70

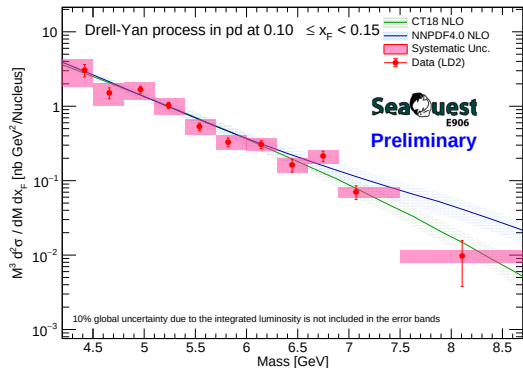


# CrossSection LD2 $0.10 \leq x_F < 0.15$ (Centroid)

## RS67

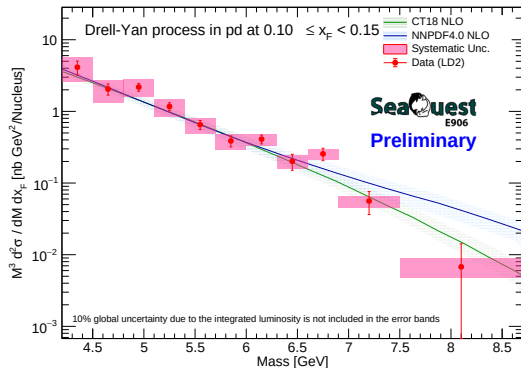


## RS57-70

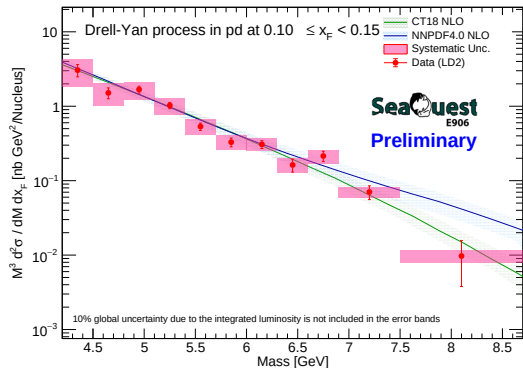


# CrossSection LD2 $0.10 \leq x_F < 0.15$ (GeoCenter)

## RS67

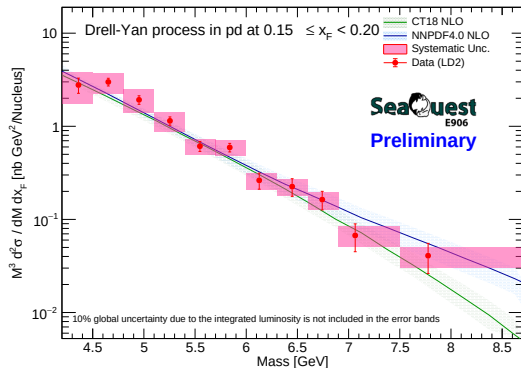


## RS57-70

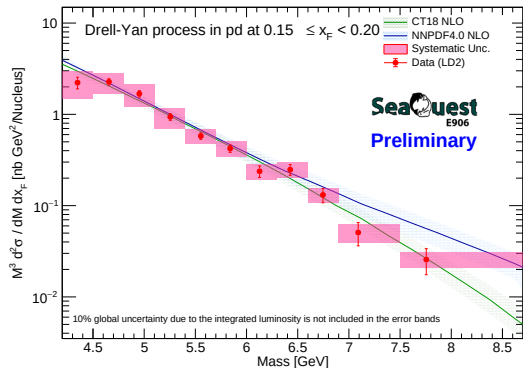


# CrossSection LD2 $0.15 \leq x_F < 0.20$ (Centroid)

## RS67

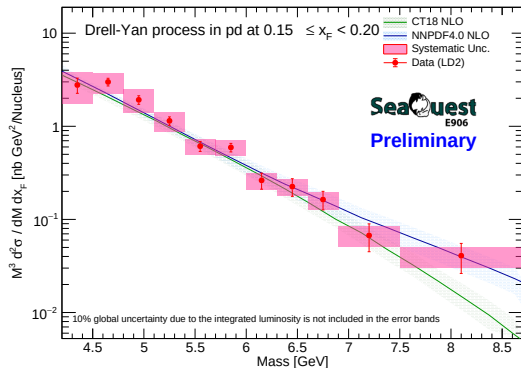


## RS57-70

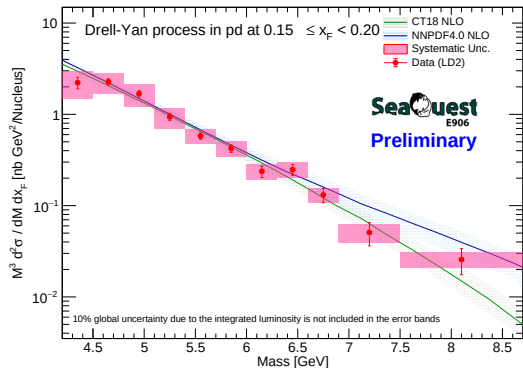


# CrossSection LD2 $0.15 \leq x_F < 0.20$ (GeoCenter)

## RS67



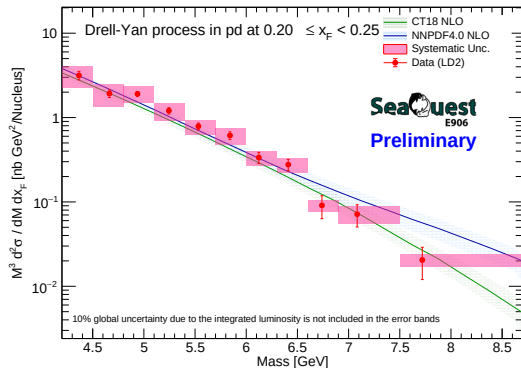
## RS57-70



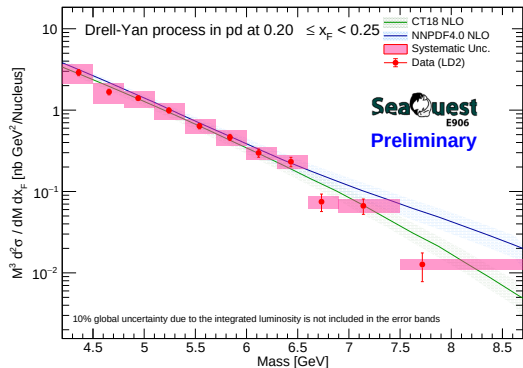


# CrossSection LD2 $0.20 \leq x_F < 0.25$ (Centroid)

## RS67

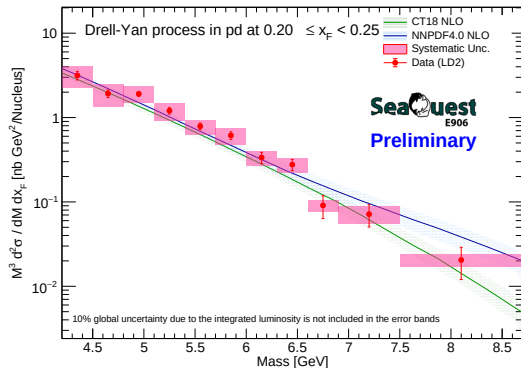


## RS57-70

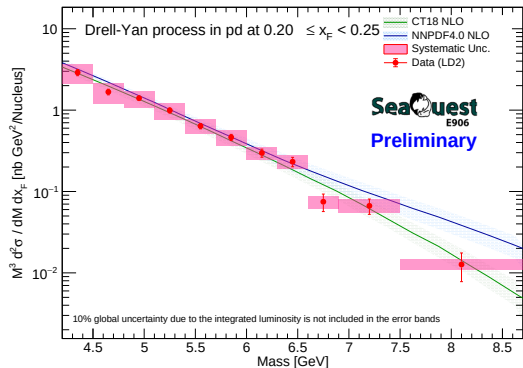


# CrossSection LD2 $0.20 \leq x_F < 0.25$ (GeoCenter)

## RS67

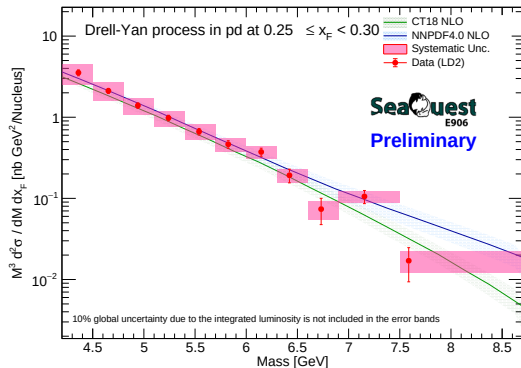


## RS57-70

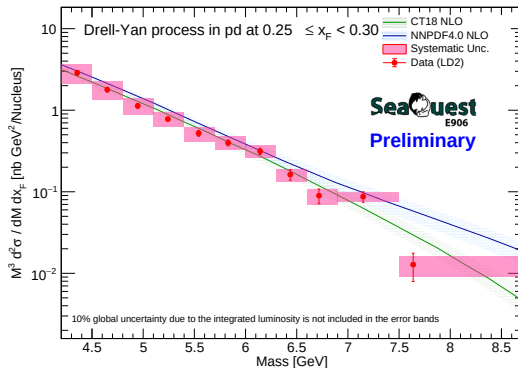


# CrossSection LD2 $0.25 \leq x_F < 0.30$ (Centroid)

## RS67

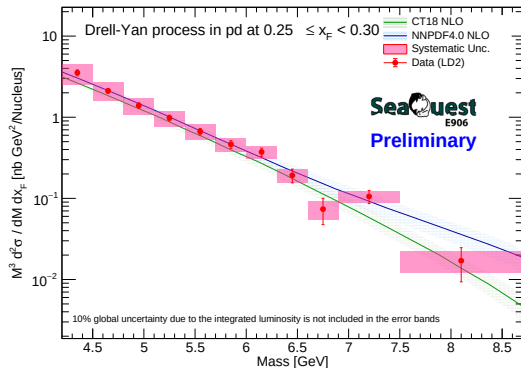


## RS57-70

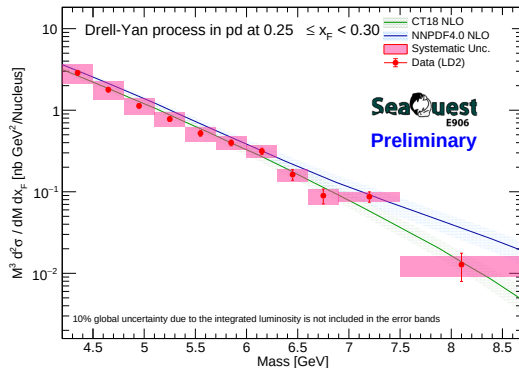


# CrossSection LD2 $0.25 \leq x_F < 0.30$ (GeoCenter)

## RS67

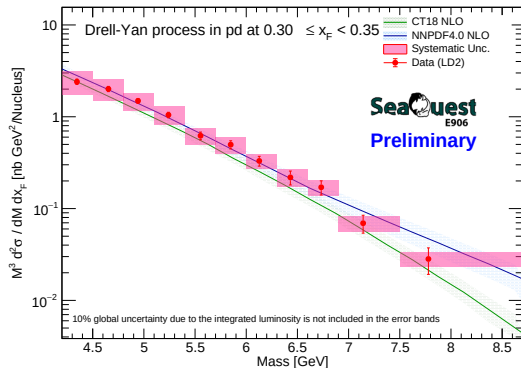


## RS57-70

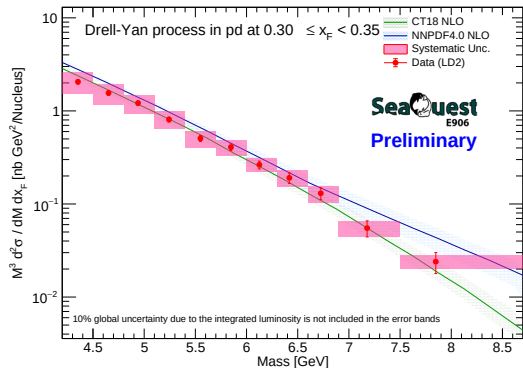


# CrossSection LD2 $0.30 \leq x_F < 0.35$ (Centroid)

## RS67

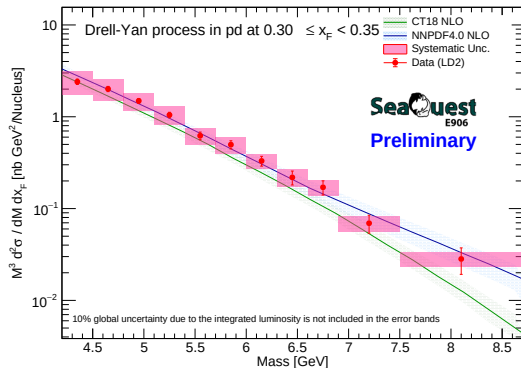


## RS57-70

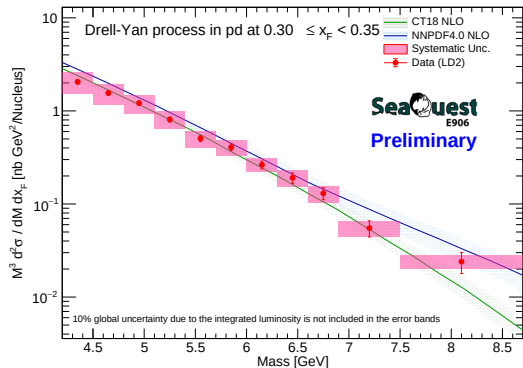


# CrossSection LD2 $0.30 \leq x_F < 0.35$ (GeoCenter)

## RS67

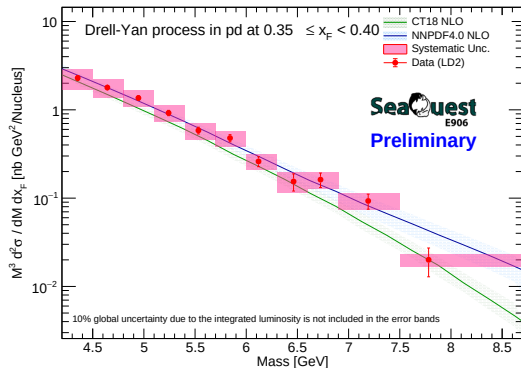


## RS57-70

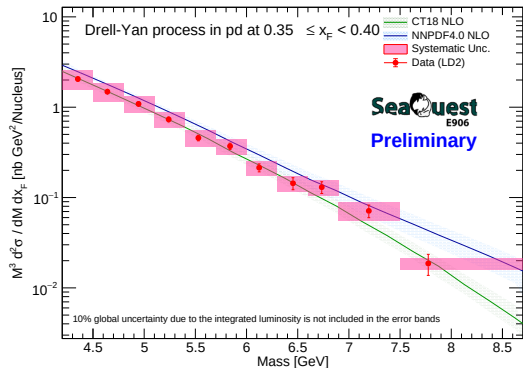


# CrossSection LD2 $0.35 \leq x_F < 0.40$ (Centroid)

## RS67

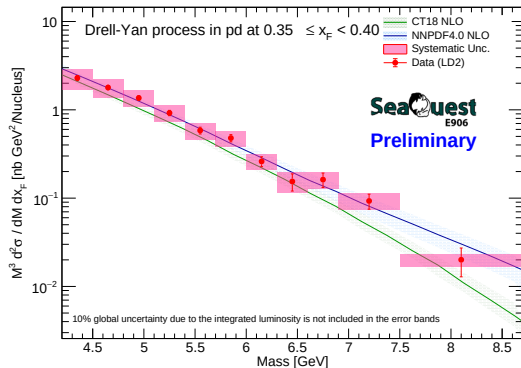


## RS57-70

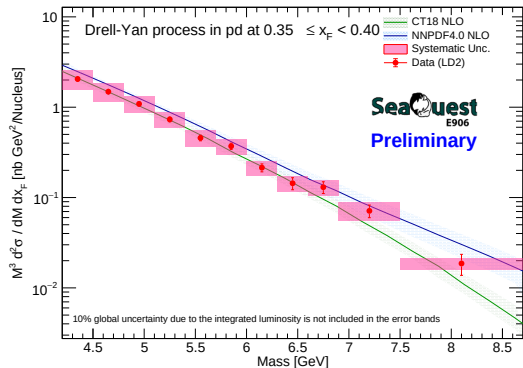


# CrossSection LD2 $0.35 \leq x_F < 0.40$ (GeoCenter)

## RS67



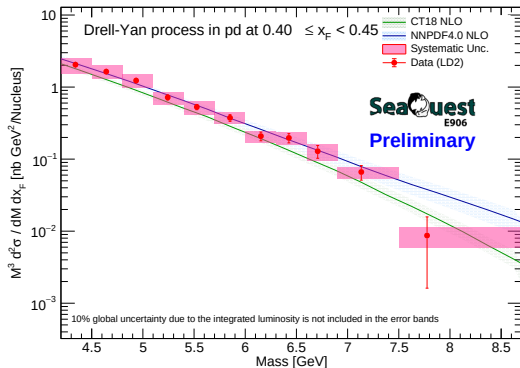
## RS57-70



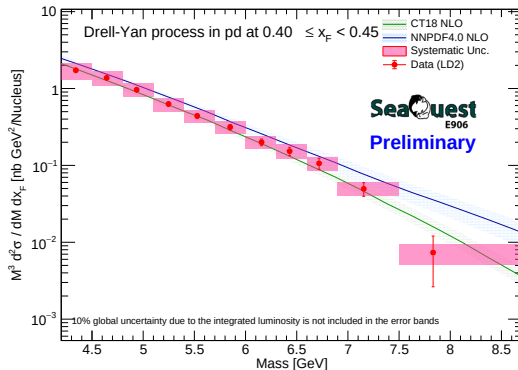


# CrossSection LD2 $0.40 \leq x_F < 0.45$ (Centroid)

## RS67

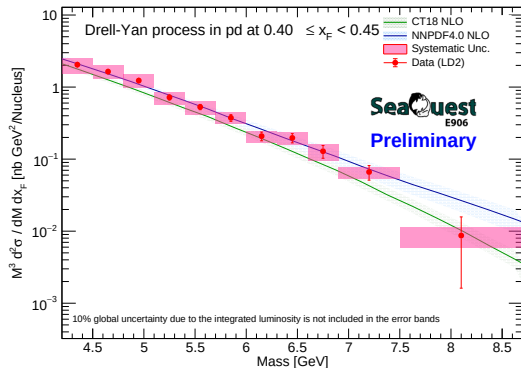


## RS57-70

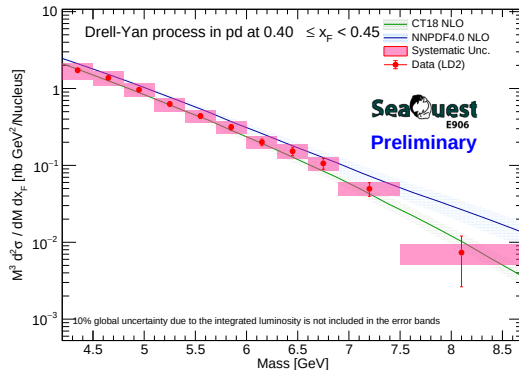


# CrossSection LD2 $0.40 \leq x_F < 0.45$ (GeoCenter)

## RS67

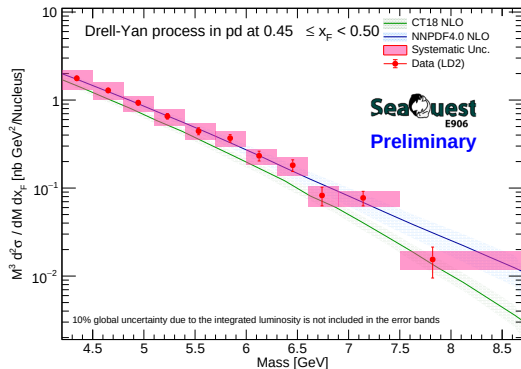


## RS57-70

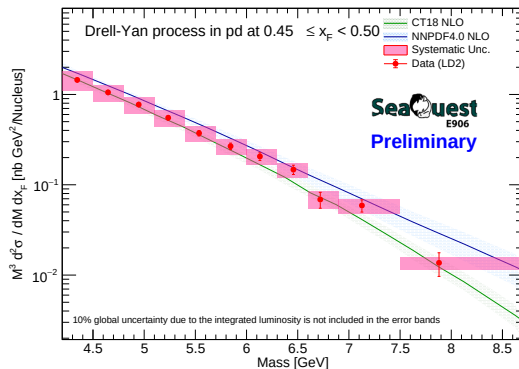


# CrossSection LD2 $0.45 \leq x_F < 0.50$ (Centroid)

## RS67

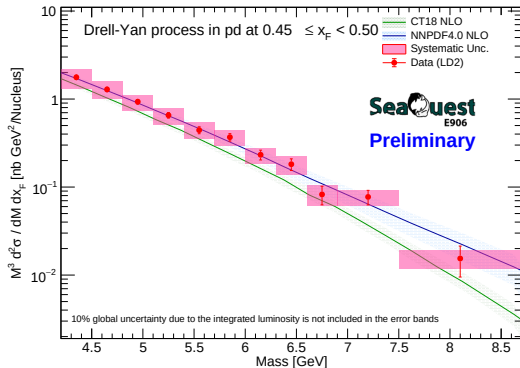


## RS57-70

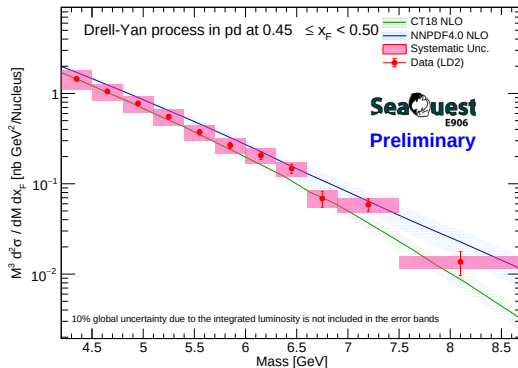


# CrossSection LD2 $0.45 \leq x_F < 0.50$ (GeoCenter)

## RS67

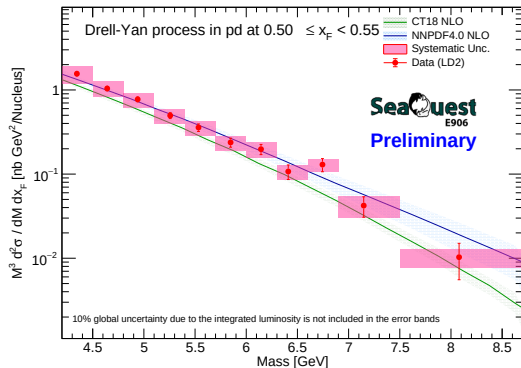


## RS57-70

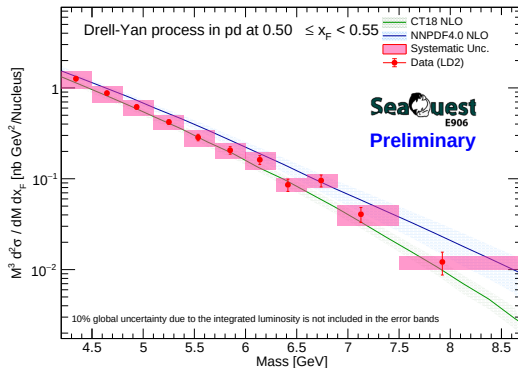


# CrossSection LD2 $0.50 \leq x_F < 0.55$ (Centroid)

## RS67

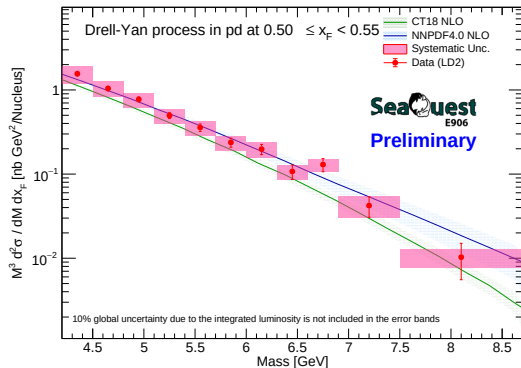


## RS57-70

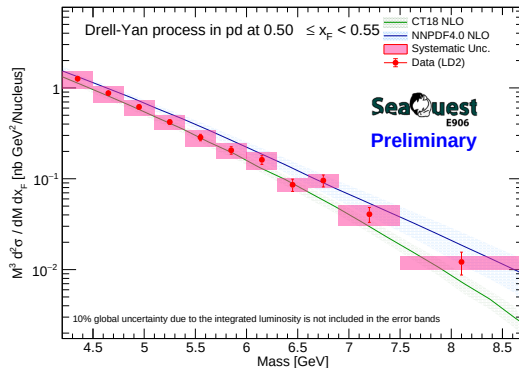


# CrossSection LD2 $0.50 \leq x_F < 0.55$ (GeoCenter)

## RS67

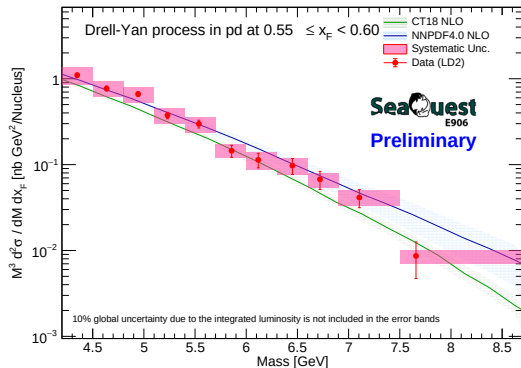


## RS57-70

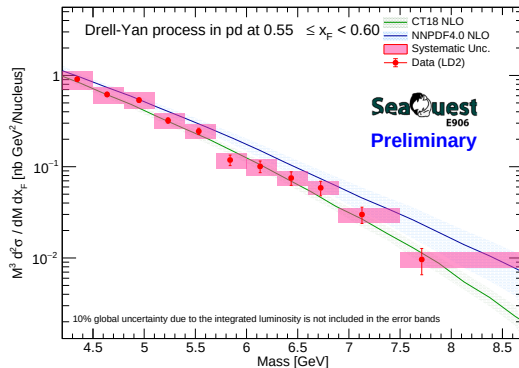


# CrossSection LD2 $0.55 \leq x_F < 0.60$ (Centroid)

## RS67

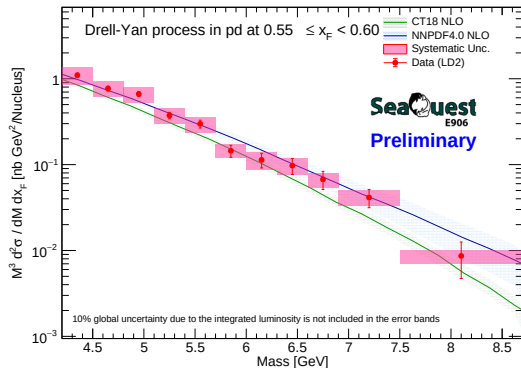


## RS57-70

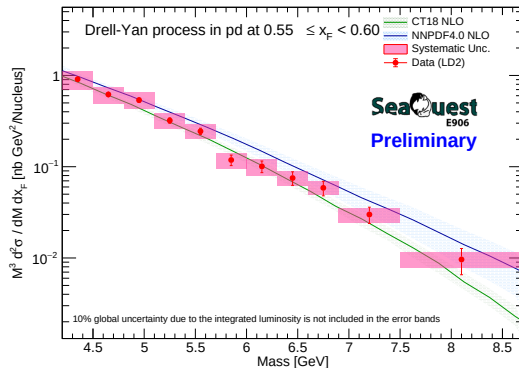


# CrossSection LD2 $0.55 \leq x_F < 0.60$ (GeoCenter)

## RS67



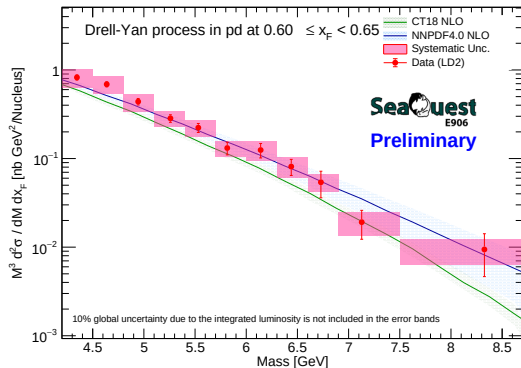
## RS57-70



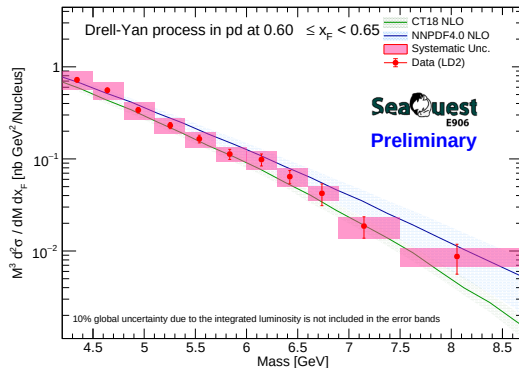


# CrossSection LD2 $0.60 \leq x_F < 0.65$ (Centroid)

## RS67

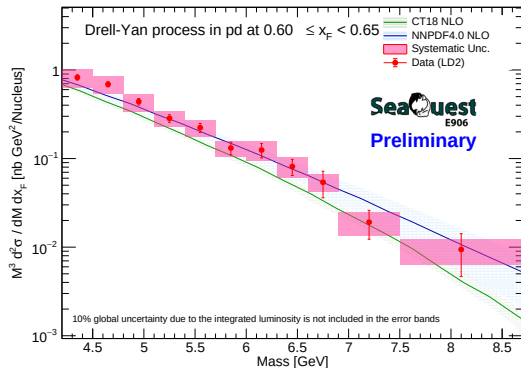


## RS57-70

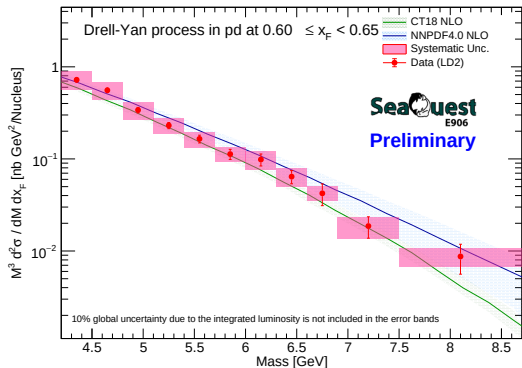


# CrossSection LD2 $0.60 \leq x_F < 0.65$ (GeoCenter)

## RS67

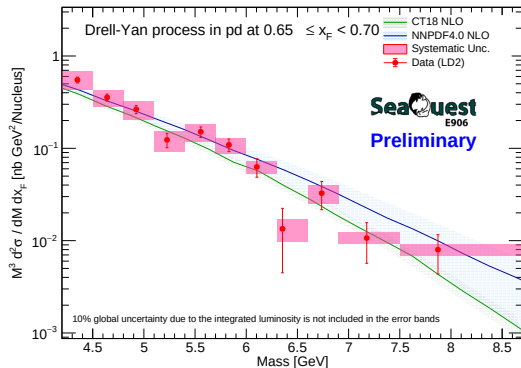


## RS57-70

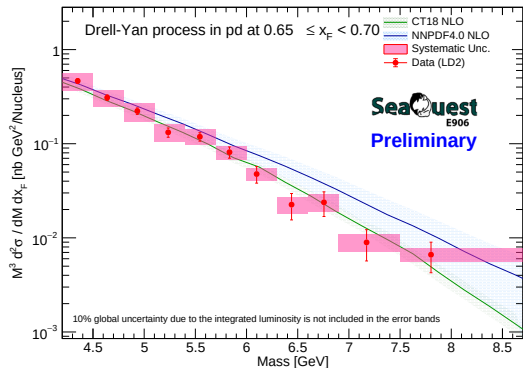


# CrossSection LD2 $0.65 \leq x_F < 0.70$ (Centroid)

## RS67

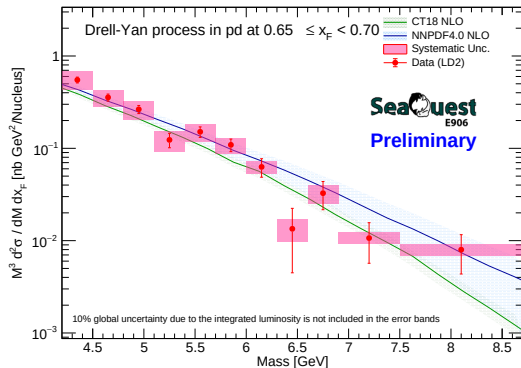


## RS57-70

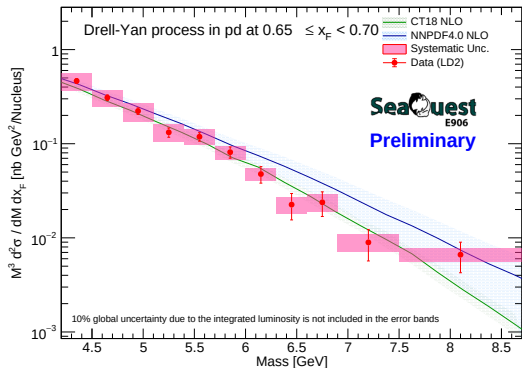


# CrossSection LD2 $0.65 \leq x_F < 0.70$ (GeoCenter)

## RS67

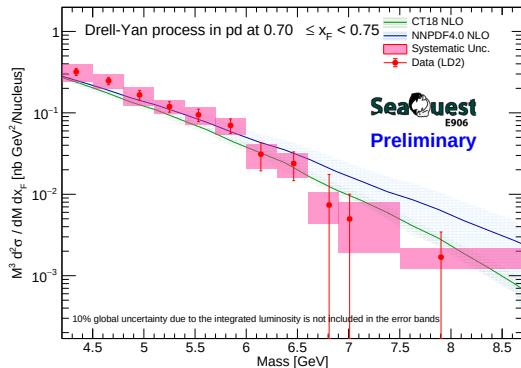


## RS57-70

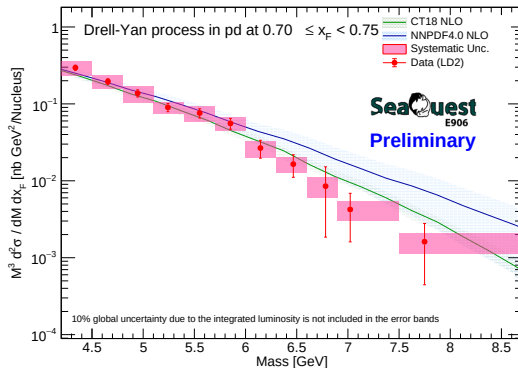


# CrossSection LD2 $0.70 \leq x_F < 0.75$ (Centroid)

## RS67

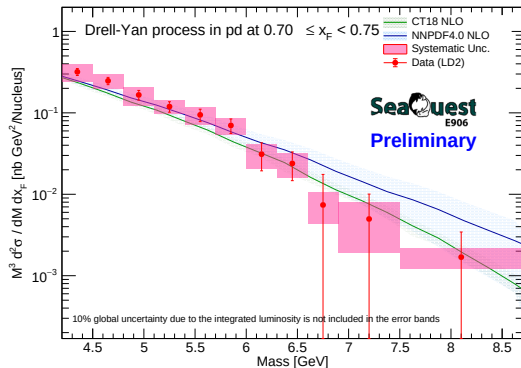


## RS57-70

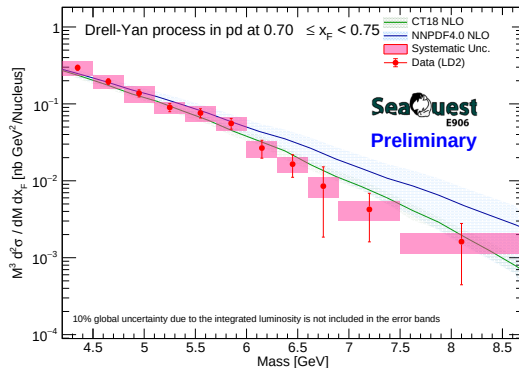


# CrossSection LD2 $0.70 \leq x_F < 0.75$ (GeoCenter)

## RS67

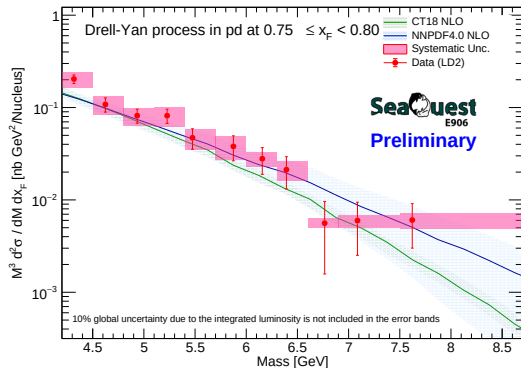


## RS57-70

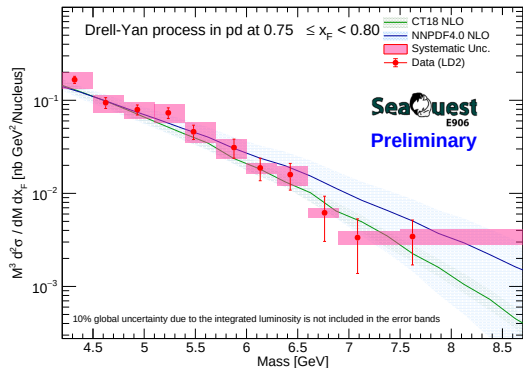


# CrossSection LD2 $0.75 \leq x_F < 0.80$ (Centroid)

## RS67

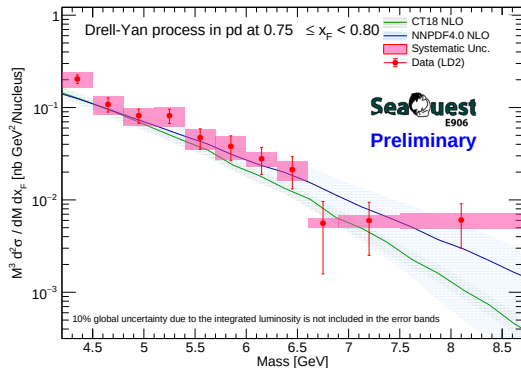


## RS57-70



# CrossSection LD2 $0.75 \leq x_F < 0.80$ (GeoCenter)

## RS67



## RS57-70

